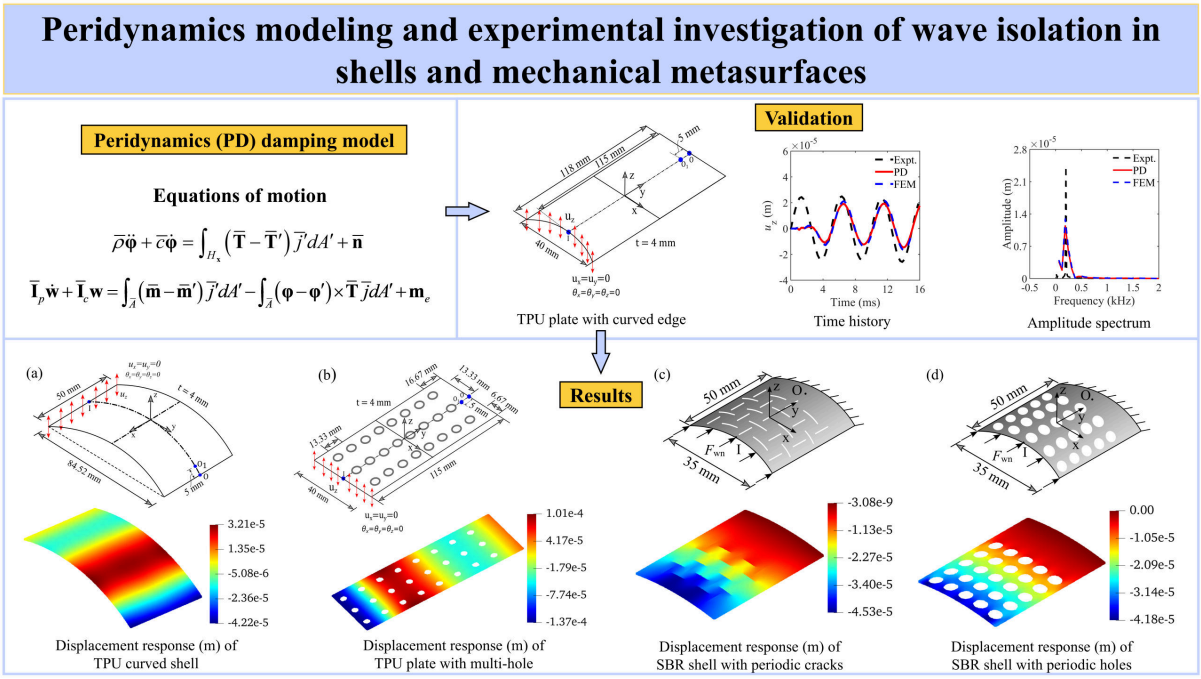
Graphical Abstract

Peridynamics modeling and experimental investigation of wave isolation in shells and mechanical metasurfaces

Nilesh Choudhary[a], Tanuj Gupta[b], Ankur Dwivedi[b], Bijaya Bikram Samal[b], Pranesh Roy[a]*, Rahul Bhartiya[a], and Bishakh Bhattacharya[b]



Highlights

Peridynamics modeling and experimental investigation of wave isolation in shells and mechanical metasurfaces

Nilesh Choudhary[a], Tanuj Gupta[b], Ankur Dwivedi[b], Bijaya Bikram Samal[b], Pranesh Roy[a*], Rahul Bhartiya[a], and Bishakh Bhattacharya[b]

- Finite deformation peridynamics (PD) shell model with damping is formulated.
- Energy dissipation is captured through damping.
- PD shell model is validated against finite element method and experiments.
- PD simulations show wave isolation in metasurfaces with periodic holes and cracks.

Peridynamics modeling and experimental investigation of wave isolation in shells and mechanical metasurfaces

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Abstract

A finite deformation Simo-Reissner peridynamics (PD) shell theory incorporating damping is developed to investigate wave isolation in shells and mechanical metasurfaces. Damping is introduced in the three-dimensional (3D) governing equations of non-ordinary state-based PD theory for modeling energy dissipation. The idea here is to employ balance of linear and angular momenta, and exploit Simo's assumption on the deformation field, to obtain the governing equations of motion of shell in the presence of damping. These damped dynamic governing equations are solved using the implicit Newmark-beta method in conjunction with the Newton-Raphson method at each time step. The displacement response obtained from the proposed PD model is validated against finite element simulations and experimental observations on thermoplastic polyurethane (TPU 95A) plates and shells subjected to harmonic excitation, both in time and frequency domains. Further, numerical investigations are carried out on mechanical metasurfaces, viz., TPU plates with periodically arranged holes and cracks, and cylindrical shells made up of styrene-butadiene rubber (SBR) with periodically arranged cracks and holes. These simulations reveal significant wave attenuation and low transmittance due to large deformation of the thin ligaments between holes and wave reflections from the hole and crack surfaces, demonstrating the efficacy of the proposed approach.

Keywords: Peridynamics; Shell; Metasurfaces; Wave isolation; Crack; Damping.

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1. Introduction

Many naturally occurring and engineered three-dimensional (3D) structures are curved surface-like where the thickness dimension is small compared to the length and width. The analysis of these structures may render significant computational advantage if the thickness dimension is homogenized or integrated out. These curved surface structures, known as shells, are widely adopted across a variety of engineering disciplines. They play a crucial role in aerospace, civil, and mechanical engineering, appear in applications such as aircraft and rocket

components, lightweight outer body of high-speed trains, architectural elements like domes and roofs, and even in biological structures such as arteries and membranes. When subjected to dynamic forces, viz., strong aerodynamic load in aircraft and high-speed trains, and blast and seismic loading on domes, wave propagates through shell which often lead to the initiation and growth of cracks. Understanding wave propagation, damage, and fracture in shell structures is therefore essential for ensuring structural integrity, safety, and long-term durability. Beyond analysis of wave propagation, modern design strategies focus on tailoring shell structures to absorb or divert wave energy, with a view to protecting important regions in the shell structure, such as regions where low-profile electronic equipment are embedded to the aerofoil or aircraft body. This has motivated the integration of shells with micro-architected mechanical metamaterials. These materials with internal structures show unconventional properties that are not typically observed in traditional materials. These properties include shape adaptability, negative Poisson's ratio, high strength-to-weight ratio, energy absorption capacity, and bandgaps. Such characteristics make shell-based metamaterials, henceforth referred to as mechanical metasurfaces, particularly suitable for applications in mechanical filtering, waveguiding, and vibration control.

Several researchers investigated the performance of various types of mechanical metasurfaces and opened new opportunities in wave manipulation. For instance, Sakovsky and Ermanni [1] employed 3D printing technique to fabricate lightweight thin shells and demonstrated their ability to block vibration-induced sound waves. A cylindrical shell model exhibiting bidirectional negative stiffness is proposed by Ren et al. [2]. Martin et al. [3] developed fluid-filled shells for acoustic metamaterial devices. Wang et al. [4] combined genetic algorithms with machine learning to perform inverse design of shell metamaterials. Yu et al. [5] analyzed the vibration transmission of double-layer cylindrical shells with fully immersed elastic connections through experimental and analytical methods. Xu et al. [6] developed an improved transfer matrix method for vibro-acoustic analysis of submerged cylindrical shells under hydrostatic pressure. Although these studies mainly focus on design and applications, shell models capable of capturing large deformations and crack propagation in the presence of energy dissipation caused by damping are still limited.

Shells are special 3D geometries where thickness is much smaller than the other dimensions. Full-fledged 3D analysis of these structures requires a large number of degrees of freedom and often leads to numerical ill-conditioning [7]. To reduce computational cost, shells are usually modeled in a locally two-dimensional (2D) manner by homogenizing the thickness dimension. Several classical shell theories exist, each offering a certain degree of accuracy. For instance, the Kirchhoff-Love theory assumes that the director vector remains normal to the deformed base surface. While suitable for thin shells, it neglects the transverse shear deformation. In contrast, the Reissner-Mindlin theory accounts for shear deformation, and thus applies to both thin and thick shells, with the director vector no longer constrained to remain normal to the deformed base surface. Influenced by the Cosserat continuum theory, finite-deformation shell models were developed by Argyris and Scharpf [8], Green and Naghdi [9], and Antman [10, 11] which paved the way for further development. A significant advancement was made by Simo and Fox [12], who introduced a geometrically exact shell model incorporating shear deformation similar to the Reissner-Mindlin theory. This model is henceforth referred to as Simo-Reissner theory. They also provided variational formulation and finite element discretization of their theory [13] without explicitly using the Christoffel symbol and thereby offered computational simplicity. The Simo-Reissner framework provides shell governing equations and rotation updates without drilling degree of freedom. However, like other classical continuum mechanics (CCM) based shell models, its equations are expressed in partial differential form,

which are undefined at the discontinuities such as cracks.

To accommodate discontinuity in continuum mechanics setup, Silling [14] introduced a fundamentally different approach known as peridynamics (PD), a nonlocal continuum theory, which reformulates continuum mechanics using integro-differential equations that remain valid at discontinuities. The initial model, known as bond-based PD (BB-PD) theory, envisions material points interacting through bonds. However, there is a restriction of fixed Poisson's ratio (0.25 for 3D/plane strain and 1/3 for plane stress) in this model that limits its versatility. To overcome this limitation, Silling et al. [15] developed state-based PD that includes ordinary state-based PD (OSB-PD) and non-ordinary state-based PD (NOSB-PD). In NOSB-PD theory, bond force depends on the collective deformation of all bonds within a material point's horizon, with no constraint to remain in the direction of the deformed bond vector, making it the most versatile PD formulation. Classical constitutive laws can be embedded in PD through the constitutive correspondence approach, allowing NOSB-PD to take advantage of the already available rich material library of CCM. Though correspondence material models suffer from zero-energy deformation modes leading to unphysical oscillations in the response, several zero-energy elimination techniques have been proposed [15–19]. Comprehensive reviews of PD theory and its developments can be found in the literature [20–22].

PD has since been applied to a wide range of problems, including wave and crack propagation, microstructural effects, material nonlinearity [25, 29], and dimensionally reduced structures. For example, Mahadeshwar et al. [23] developed a finite deformation PD shell theory based on Simo-Reissner-Mindlin model and studied wave propagation through metasurfaces with a view to enhance bandgaps. O'Grady and Foster [24] simulated torsion-induced fracture of plates and flat shells using NOSB-PD without transverse shear effects. Nguyen and Oterkus [26] developed an OSB-PD shell model and applied it to stiffened shell and thermomechanical fracture. Kumar et al. [27] proposed a PD viscoelastic shell theory to capture creep deformation in shells and metasurfaces. Jiang et al. [28] introduced an extended BB-PD shell model for failure of cylindrical shells under hydrostatic pressure. Xia et al. [30] employed an isogeometric PD theory to predict crack propagation in Reissner-Mindlin shells. Sajal and Roy [31] applied NOSB-PD theory to simulate wave propagation through locally resonant metamaterials and demonstrated their energy absorption capacity. Dey et al. [36] presented a PD multi-shell theory to model crack and wave propagation in complex multi-shell metasurfaces. Despite these advances, existing PD shell formulations primarily address elastic response, with limited attention to damping and experimental validation. Since damping is critical for accurately predicting wave propagation through materials, especially polymers, experiments are necessary to validate computational predictions, an aspect that is still lacking in existing literature, to the best of the authors' knowledge.

The present work investigates dynamic response of shells and mechanical metasurfaces under harmonic and irregular excitations. Geometric imperfections such as holes and cracks are introduced in shells to manipulate wave propagation, enabling attenuation and wave isolation. While periodically arranged holes with slender cell walls induce large deformation, and often, elastic instabilities, that absorb wave energy, periodically arranged cracks act as wave barriers that reflect incident waves. As polymeric plates and shells are used, damping is included in the model for realistic representation of energy dissipation. Note that plate is a special case of shell with zero curvature.

In this work, we develop a finite deformation Simo-Reissner NOSB-PD shell theory with damping to study wave propagation and wave isolation in shells and mechanical metasurfaces. The shell equations of motion with damping are derived from 3D PD balance of linear and angular momenta employing Simo-Reissner kinematic assumption and through-the-thickness

integration. Constitutive correspondence method is used to incorporate classical shell material models in PD setup. The Newmark-beta method is used to numerically solve the shell equations of motion. PD simulations and experiments are performed on thermoplastic polyurethane (TPU 95A) solid plates and shells under harmonic excitation. The PD results are compared with finite element method (FEM) solutions obtained using ABAQUS® and experimental results using 3D Laser Doppler Vibrometer, which show good agreement. Further, TPU plates and styrene-butadiene rubber (SBR) shells with periodic holes and cracks subjected to dynamic loading are simulated which demonstrate significant wave attenuation and large bandgap.

2. Overview of PD theory and classical theory of shells

This section briefly describes the PD theory and the classical theory of shells. In PD theory, the discussion covers kinematics, the equations of motion, constitutive correspondence, and the control of zero-energy modes. For the classical shell theory, the focus is on the Simo-Reissner deformation approximation and the shell equations of motion.

2.1. PD theory

PD is a nonlocal continuum theory introduced by Silling [14] and Silling et al. [15] as a reformulation of CCM. In CCM, governing equations are expressed in terms of partial derivatives, which assume that material points interact only with their immediate neighbors through direct contact. This local interaction assumption breaks down in the presence of discontinuities such as cracks and voids as CCM predicts stress singularity at sharp crack tips and partial derivative terms in the governing equations are not defined at discontinuities. In contrast, the governing equations of PD are integro-differential, which are well-defined in regions containing discontinuities. The NOSB-PD formulation accounts for nonlocal interaction among material points, where the interaction for a point is defined over a finite distance, known as the horizon which has a radius δ.

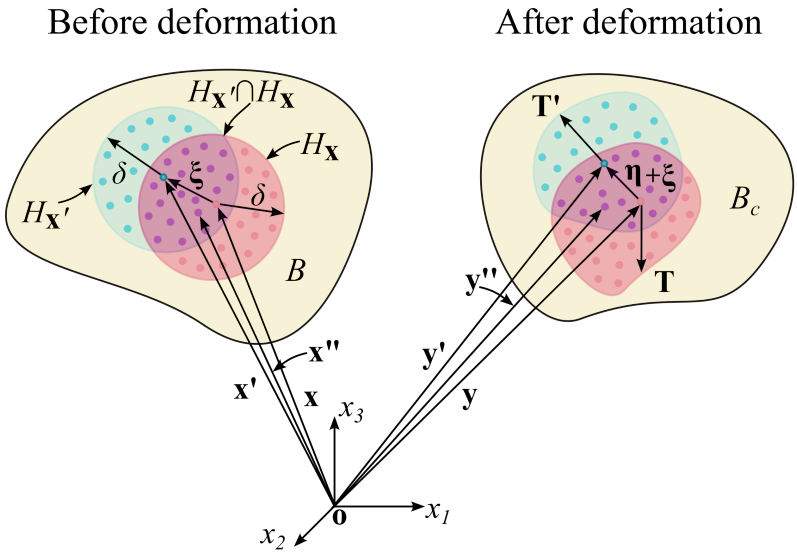


Fig. 1: Reference (B) and current configurations (B_c) of PD showing interactions between material points. A point $\mathbf{x}$ interacts with a neighboring point $\mathbf{x}'$ located within its horizon $H_{\mathbf{x}}$, defined by a radius δ. Under the application of external forces, the points displaced by $\mathbf{u}$ and $\mathbf{u}'$, and moved to new positions $\mathbf{y}$ and $\mathbf{y}'$ in the deformed state.

Consider the reference and current configurations of a material body as shown in Fig. 1. The material point $\mathbf{x}$ interacts with another material point $\mathbf{x}'$ within the horizon $H_{\mathbf{x}}$. The relative

position vector between these two points in the reference configuration is called the bond vector, expressed as $\boldsymbol{\xi} = \mathbf{x}' - \mathbf{x}$. When the body undergoes deformation, these material points move to new positions $\mathbf{y}$ and $\mathbf{y}'$, due to displacement vectors $\mathbf{u}$ and $\mathbf{u}'$, respectively. The relative displacement vector between these two points is denoted by $\boldsymbol{\eta} = \mathbf{u}' - \mathbf{u}$ and the deformed bond vector can be given as $\mathbf{y}' - \mathbf{y} = \mathbf{u}' - \mathbf{u} + \mathbf{x}' - \mathbf{x} = \boldsymbol{\eta} + \boldsymbol{\xi}$. The governing equation of motion of PD is given as:

$$\rho\,\ddot{\mathbf{u}}(\mathbf{x},t) = \int_{H_{\mathbf{x}}} \left(\mathbf{T}[\mathbf{x},t]\langle\boldsymbol{\xi}\rangle - \mathbf{T}[\mathbf{x}',t]\langle-\boldsymbol{\xi}\rangle\right) dV_{\mathbf{x}'} + \rho\,\mathbf{b}(\mathbf{x},t). \tag{1}$$

Here $\mathbf{T}$ denotes the internal force density, and $\rho\,\mathbf{b}$ represents the body force density, and $\rho\,\ddot{\mathbf{u}}$ corresponds to the inertial pseudo-force density with ρ the mass density, expressed as the product of mass density and acceleration, which is a function of material point $\mathbf{x}$ and time t.

For constitutive correspondence method [15], one needs to define the nonlocal deformation gradient tensor $\bar{\mathbf{F}}$ as:

$$\bar{\mathbf{F}} := \int_{H_{\mathbf{x}}} w\langle\boldsymbol{\xi}\rangle(\mathbf{y}' - \mathbf{y}) \otimes \boldsymbol{\xi}\, dV_{\mathbf{x}'}\, \mathbf{K}^{-1}, \tag{2}$$

where $w\langle\boldsymbol{\xi}\rangle = \left(\frac{\delta}{\|\boldsymbol{\xi}\|}\right)^2$ [32, 33], $w\langle\boldsymbol{\xi}\rangle$ denotes the influence function which assigns weight to the interaction between two material points based on their relative distance, $\mathrm{d}V_{\mathbf{x}'}$ is the infinitesimally small volume associated with the material point $\mathbf{x}'$ and the term $\mathbf{K} = \int_{H_{\mathbf{x}}} w\langle\boldsymbol{\xi}\rangle\, \boldsymbol{\xi} \otimes \boldsymbol{\xi}\, dV_{\mathbf{x}'}$ is known as shape tensor.

In the constitutive correspondence approach, unphysical oscillations occur in the solution which is known as zero-energy modes and the integral definition of $\bar{\mathbf{F}}$ is responsible for this issue, as mentioned by Gu et al. [35]. Different methods are suggested to control the zero-energy modes. The method proposed by Chen [17], and Chen and Spencer [18] is considered in this article where the nonlocal deformation gradient tensor $\hat{\mathbf{F}}$ is calculated over the intersected horizons of two interacting material points $\mathbf{x}$ and $\mathbf{x}'$ as follows:

$$\hat{\mathbf{F}} := \int_{H_{\mathbf{x}} \cap H_{\mathbf{x}'}} w\langle\boldsymbol{\zeta}\rangle(\mathbf{y}'' - \mathbf{y}) \otimes \boldsymbol{\zeta}\, dV_{\mathbf{x}''}\, \hat{\mathbf{K}}^{-1}. \tag{3}$$

Here $\boldsymbol{\zeta} = \mathbf{x}'' - \mathbf{x}$ and $\hat{\mathbf{K}} = \int_{H_{\mathbf{x}} \cap H_{\mathbf{x}'}} w\langle\boldsymbol{\zeta}\rangle\, \boldsymbol{\zeta} \otimes \boldsymbol{\zeta}\, \mathrm{d}V_{\mathbf{x}''}$ is the two-point shape tensor. Note that a material point within $H_{\mathbf{x}} \cap H_{\mathbf{x}'}$ is denoted as $\mathbf{x}''$, and its deformed position is given by $\mathbf{y}''$.

The zero-energy corrected force density vector $\mathbf{T}$ can be given as

$$\mathbf{T}\langle\boldsymbol{\xi}\rangle = w\langle\boldsymbol{\xi}\rangle\, \mathbf{P}(\hat{\mathbf{F}})\, \mathbf{K}^{-1}\boldsymbol{\xi}, \tag{4}$$

where $\mathbf{P}$ is the first Piola-Kirchhoff stress tensor.

2.2. Classical Simo-Reissner theory of shells

A shell is a structural element characterized by the thickness dimension much smaller than its other dimensions. Shell can be modeled as a locally 2D body by integrating through-the-thickness. This dimensional reduction significantly decreases the computational cost in numerical simulations while preserving the important features, including membrane, bending, and shear responses. According to Simo and Fox [12], the position vector of a material point in the current configuration of the shell is expressed as

$$\Phi(z^1, z^2, z, t) = \boldsymbol{\phi}(z^1, z^2, t) + z\,\mathbf{t}(z^1, z^2, t), \tag{5}$$

where z denotes the coordinate along the thickness direction and Φ represents the position vector of a point in the 3D body. As shown in Figs. 2(a) and 2(b), curvilinear coordinates z^1 and z^2 describe the shell base surface, and $\boldsymbol{\phi}$ is the position vector of a point in the base surface in the current configuration at $z = 0$.

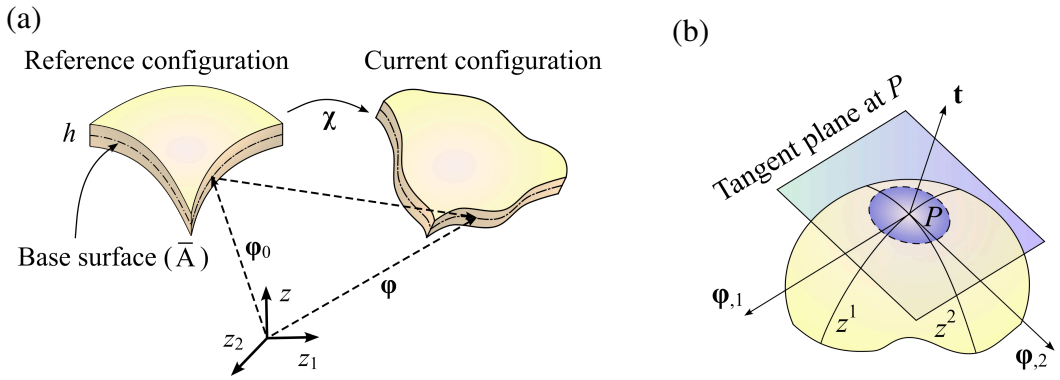


Fig. 2: Geometry of a shell structure: (a) reference and current configurations of the shell, (b) locally 2D tangent plane at a point P in the current configuration.

Simo and Fox [12] proposed a five degrees of freedom shell model by omitting the drilling rotation. The shell equations of motion can be written as:

$$\bar{\rho}\,\ddot{\boldsymbol{\phi}} = \frac{1}{\bar{j}}\left(\bar{j}\mathbf{n}^{\alpha}\right)_{,\alpha} + \bar{\mathbf{n}}, \tag{6}$$

$$\bar{I}_p\,\dot{\mathbf{w}} = \frac{1}{\bar{j}}\left(\bar{j}\mathbf{m}^{\alpha}\right)_{,\alpha} + \boldsymbol{\phi}_{,\alpha} \times \mathbf{n}^{\alpha} + \mathbf{m}_e. \tag{7}$$

Here $\bar{\rho}$ denotes the surface mass density and $\bar{I}_p$ represents the surface moment of inertia. The vectors $\mathbf{n}^{\alpha}$ and $\mathbf{m}^{\alpha}$ correspond to the stress resultant and stress couple, respectively. The terms $\bar{\mathbf{n}}$ and $\mathbf{m}_e$ represent distributed body force and body moment per unit area, and $\bar{j}$ denotes the square root of the determinant of the surface metric tensor.

3. Finite deformation Simo-Reissner PD shell theory with damping

We propose the PD governing equation of motion [23] with damping as

$$\rho\,\ddot{\Phi} + c\,\dot{\Phi} = \int_{H_{\mathbf{x}}} (\mathbf{T} - \mathbf{T}')\,dV' + \rho\,\mathbf{b}, \tag{8}$$

where c is the damping coefficient. c is analogous to the viscous damping coefficient in a single degree of freedom system, where it can be expressed in terms of damping factor ξ and natural frequency ω_n as

$$c = 2\xi\,\omega_n\rho = \alpha\rho, \tag{9}$$

where α is the Rayleigh damping coefficient given by $\alpha = 2\xi\,\omega_n$. The value of α can be approximately determined in case of 3D structures by estimating ω_n. A shell has infinite degrees of freedom as it is a continuous system. Also, after discretization into particles, the shell becomes a multi-degree-of-freedom system. Therefore, there are multiple natural frequencies and damping coefficients. Note that the damping coefficients are higher for the higher modes. As we are considering Rayleigh damping, the damping matrix is dependent on natural frequencies.

As experimental determination of all the damping coefficients is considerably difficult, therefore, in the present work, we considered a conservative estimate of the damping coefficient so that the experiment can be conducted easily. The damping matrix is taken as a diagonal matrix which is calculated as damping coefficient times the identity matrix. We roughly estimated ω_n using the following formula based on single-degree-of-freedom system as $\omega_n = \sqrt{k/m}$, where k is the stiffness and m is the mass of the shell. The value of k is calculated from the axial stiffness expression $k = AE/L$, where A is the cross-sectional area and L is the length of the shell in the direction of wave propagation. m can be determined from the relation: $m = \rho V$, where V is the volume of the shell.

Integrating Eq. (8) over the shell body B and extending $H_{\mathbf{x}}$ to B as there is no interaction outside the horizon, we may write

$$\int_B \rho\, \ddot{\Phi}\, dV + \int_B c\dot{\Phi}\, dV = \int_B \int_B \left(\mathbf{T} - \mathbf{T}'\right) dV'\, dV + \int_B \rho\, \mathbf{b}\, dV. \tag{10}$$

Note that Eq. (10) represents the balance of linear momentum for 3D body of shell. Integrating the damping term in Eq. (10), through the shell thickness, one obtains

$$\int_B c\dot{\Phi}\, dV = \int_{\bar{A}} \int_{h^-}^{h^+} c\, \left(\dot{\boldsymbol{\phi}} + z\dot{\mathbf{t}}\right) j\, dz\, dA = \int_{\bar{A}} \bar{c}\, \bar{j}\, \dot{\boldsymbol{\phi}}\, dA, \tag{11}$$

where $\bar{c} = \frac{1}{\bar{j}} \int_{h^-}^{h^+} c\, j\, dz$, j represents the square root of the determinant of 3D metric tensor, $\bar{A}$ is the base surface, and $z \in [h^-, h^+]$. Using the approximation $j \approx \bar{j}$ (Kumar et al. [27]), an approximate value of $\bar{c}$ can be obtained as $\bar{c} \approx ch = \alpha\rho h$, where h is the shell thickness. For the simplification of other terms in Eq. (11), one may refer to Mahadeshwar et al.[23].

Employing the Simo-Reissner assumption on deformation field (Eq. (5)) and integrating through-the-thickness [23], the first shell equation of motion with damping can be obtained as

$$\bar{\rho}\, \ddot{\boldsymbol{\phi}} + \bar{c}\, \dot{\boldsymbol{\phi}} = \int_{H_{\mathbf{x}}} \left(\bar{\mathbf{T}} - \bar{\mathbf{T}}'\right) \bar{j}'\, d\bar{A}' + \bar{\mathbf{n}}, \tag{12}$$

where $\bar{\mathbf{T}}$ and $\bar{\mathbf{T}}'$ are the resultant force state vectors and $\bar{\mathbf{n}}$ is the resultant body force vector defined as follows:

$$\bar{\mathbf{T}} = \frac{1}{\bar{j}'\bar{j}} \int_{h^-}^{h^+} \int_{h^-}^{h^+} \mathbf{T}\, j'\, j\, dz'\, dz, \tag{13}$$

$$\bar{\mathbf{T}}' = \frac{1}{\bar{j}'\bar{j}} \int_{h^-}^{h^+} \int_{h^-}^{h^+} \mathbf{T}'\, j'\, j\, dz'\, dz, \tag{14}$$

$$\bar{\mathbf{n}} = \frac{1}{\bar{j}} \int_{h^-}^{h^+} \rho\, \mathbf{b}\, j\, dz. \tag{15}$$

Next, we take recourse to the balance of angular momentum. Taking the cross product of Φ with Eq. (8) and integrating over the entire shell body yields

$$\int_B \Phi \times \rho\ddot{\Phi}\, dV + \int_B \Phi \times c\dot{\Phi}\, dV = \int_B \int_B \Phi \times (\mathbf{T} - \mathbf{T}')\, dV'\, dV + \int_B \Phi \times \rho\mathbf{b}\, dV. \tag{16}$$

Eq. (16) can be rearranged as

$$\begin{aligned}
&\int_B \Phi \times \rho\, \ddot{\Phi}\, dV + \int_B \Phi \times c\dot{\Phi}\, dV \\
&= \int_B \int_B \Big(\boldsymbol{\phi} \times (\mathbf{T} - \mathbf{T}') - (\boldsymbol{\phi} - \boldsymbol{\phi}') \times \mathbf{T} + (\Phi' - \boldsymbol{\phi}') \times \mathbf{T} \Big) dV'\, dV \\
&\quad - \int_B \int_B \Big((\Phi - \boldsymbol{\phi}) \times \mathbf{T}' \Big) dV'\, dV + \int_B \Phi \times \rho\, \mathbf{b}\, dV. \qquad (17)
\end{aligned}$$

Next, we take the cross product of $\boldsymbol{\phi}$ with Eq. (9) and integrate over the shell body to obtain

$$\int_B \boldsymbol{\phi} \times \rho \ddot{\Phi}\, dV + \int_B \boldsymbol{\phi} \times c\dot{\Phi}\, dV = \int_B \int_B \boldsymbol{\phi} \times (\mathbf{T} - \mathbf{T}')\, dV'\, dV + \int_B \boldsymbol{\phi} \times \rho \mathbf{b}\, dV. \qquad (18)$$

Subtracting Eq. (18) from Eq. (17), we may write

$$\begin{aligned}
&\int_B (\Phi - \boldsymbol{\phi}) \times \rho \ddot{\Phi}\, dV + \int_B (\Phi - \boldsymbol{\phi}) \times c\dot{\Phi}\, dV \\
&= \int_B \int_B \big(- (\boldsymbol{\phi} - \boldsymbol{\phi}') \times \mathbf{T} + (\Phi' - \boldsymbol{\phi}') \times \mathbf{T} \big) dV'\, dV - \int_B \int_B (\Phi - \boldsymbol{\phi}) \times \mathbf{T}'\, dV'\, dV + \int_B (\Phi - \boldsymbol{\phi}) \times \rho \mathbf{b}\, dV. \qquad (19)
\end{aligned}$$

The second term on the left-hand side of Eq. (19) can be simplified as

$$\int_B (\Phi - \boldsymbol{\phi}) \times c\dot{\Phi}\, dV = \int_{\bar{A}} \int_{h^-}^{h^+} z\mathbf{t} \times c \left(\dot{\boldsymbol{\phi}} + z\dot{\mathbf{t}} \right) j\, dz\, dA = \int_{\bar{A}} \int_{h^-}^{h^+} c j z^2\, dz \left(\mathbf{t} \times \dot{\mathbf{t}} \right) dA = \int_{\bar{A}} \bar{\mathbf{I}}_c \bar{j} \bar{w}\, dA. \qquad (20)$$

$\bar{\mathbf{I}}_c = \frac{1}{\bar{j}} \left(\int_{h^-}^{h^+} c j z^2\, dz \right) \mathbf{I} \approx \frac{ch^3}{12} \mathbf{I} = \frac{\alpha \rho h^3}{12} \mathbf{I}$ with $\mathbf{I}$ being the identity tensor. The other terms in Eq. (20) can be simplified following similar lines as in Mahadeshwar et al. [23].

Using the Simo-Reissner approximation on the deformation field (Eq. (5)) and homogenizing through the thickness [23], we obtain the second shell equation of motion with damping as

$$\bar{\mathbf{I}}_p \dot{\mathbf{w}} + \bar{\mathbf{I}}_c \mathbf{w} = \int_{\bar{A}} (\bar{\mathbf{m}} - \bar{\mathbf{m}}') \bar{j}'\, dA' - \int_{\bar{A}} (\boldsymbol{\phi} - \boldsymbol{\phi}') \times \bar{\mathbf{T}} \bar{j}\, dA' + \mathbf{m}_e, \qquad (21)$$

where $\bar{\mathbf{I}}_p = \frac{1}{\bar{j}} \left(\int_{h^-}^{h^+} \rho j z^2\, dz \right) \mathbf{I}$ is the surface inertia tensor, $\bar{\mathbf{I}}_c$ is the damping tensor, $\bar{\mathbf{m}}$ and $\bar{\mathbf{m}}'$ are the resultant moment vector states, and $\mathbf{m}_e$ is the resultant body moment vector defined as

$$\bar{\mathbf{m}} = \frac{1}{\bar{j}' \bar{j}} \int_{h^-}^{h^+} \int_{h^-}^{h^+} (\Phi' - \boldsymbol{\phi}') \times \mathbf{T}\, j' j\, dz'\, dz, \qquad (22)$$

$$\bar{\mathbf{m}}' = \frac{1}{\bar{j}' \bar{j}} \int_{h^-}^{h^+} \int_{h^-}^{h^+} (\Phi - \boldsymbol{\phi}) \times \mathbf{T}'\, j' j\, dz'\, dz, \qquad (23)$$

$$\mathbf{m}_e = \frac{1}{\bar{j}} \int_{h^-}^{h^+} (\Phi - \boldsymbol{\phi}) \times \rho \mathbf{b}\, j\, dz. \qquad (24)$$

4. Numerical implementation

The governing equations of the Simo-Reissner PD shell theory with damping are solved using Newmark-beta method based iterative scheme for time integration and the Newton-Raphson method for convergence at every time step. At each time step, the incremental displacement and rotation updates can be given as [23]

$$\Delta\boldsymbol{\phi}_i^{(n+1)} = \Delta\boldsymbol{\phi}_i^n + \delta\Delta\boldsymbol{\phi}_i, \tag{25}$$

$$\Delta\boldsymbol{\theta}_i^{(n+1)} = \Delta\boldsymbol{\theta}_i^n + \delta\Delta\boldsymbol{\theta}_i, \tag{26}$$

where $\Delta\boldsymbol{\phi}_i$ and $\Delta\boldsymbol{\theta}_i$ are the displacement and rotation increments at the i-th time step, and the superscripts n denotes iteration counter for the Newton-Raphson procedure.

Note that $\mathbf{t}$ denotes the director vector at a point in the deformed configuration of the shell. The change of $\mathbf{t}$ is given by $\Delta\mathbf{t} = \Delta\boldsymbol{\theta} \times \mathbf{t} = \text{skew}(\Delta\boldsymbol{\theta})\,\mathbf{t}$, where $\Delta\boldsymbol{\theta}$ is the incremental rotation pseudovector, and the symbol $\text{skew}(\Delta\boldsymbol{\theta})$ denotes the skew-symmetric matrix associated with the axial vector $\Delta\boldsymbol{\theta}$. After solving for $\Delta\boldsymbol{\theta}$, $\mathbf{t}$ can be updated as

$$\mathbf{t}_{n+1} = \exp(\Delta\mathbf{t}_n) := \cos\|\Delta\mathbf{t}_n\|\,\mathbf{t}_n + \frac{\sin\|\Delta\mathbf{t}_n\|}{\|\Delta\mathbf{t}_n\|}\,\Delta\mathbf{t}_n \tag{27}$$

The Newmark-beta method is employed for time-marching the dynamic governing equations of shell in the presence of damping. The Newton-Raphson method is used to solve the nonlinear time-discretized governing equations at every time step. Consider that the shell equations of motion are satisfied at i-th time step, i.e.,

$$\bar{\rho}\,\ddot{\boldsymbol{\phi}}_i + \bar{c}\,\dot{\boldsymbol{\phi}}_i = -(\mathbf{R}_f)_i + \bar{\mathbf{n}}_i, \tag{28}$$

$$\bar{\mathbf{I}}_p\,\ddot{\boldsymbol{\theta}}_i + \bar{\mathbf{I}}_c\,\dot{\boldsymbol{\theta}}_i = -(\mathbf{R}_m)_i + (\mathbf{m}_e)_i, \tag{29}$$

where $\mathbf{R}_f = -\int_{H_\mathbf{x}} (\bar{\mathbf{T}} - \bar{\mathbf{T}}')\,\bar{j}'\,dA'$ and $\mathbf{R}_m = -\int_{H_\mathbf{x}} (\bar{\mathbf{m}} - \bar{\mathbf{m}}')\,\bar{j}'\,dA' + \int_{H_\mathbf{x}} (\boldsymbol{\phi} - \boldsymbol{\phi}') \times \bar{\mathbf{T}}\,\bar{j}'\,dA'$. In order to proceed to the (i+1)-th time step, the position vector of the points in the base surface ($\boldsymbol{\phi}$) and the rotation pseudo-vector field ($\boldsymbol{\theta}$) are incremented with unknown vector fields $\Delta\boldsymbol{\phi}_i$ and $\Delta\boldsymbol{\theta}_i$ as

$$\boldsymbol{\phi}_{i+1} = \boldsymbol{\phi}_i + \Delta\boldsymbol{\phi}_i, \tag{30}$$

$$\boldsymbol{\theta}_{i+1} = \boldsymbol{\theta}_i + \Delta\boldsymbol{\theta}_i. \tag{31}$$

Using $\Delta\boldsymbol{\theta}_i$, the director vector $\mathbf{t}$ can be incremented as:

$$\mathbf{t}_{i+1} = \exp\big(\text{skew}(\Delta\boldsymbol{\theta}_i)\,\mathbf{t}_i\big). \tag{32}$$

With these incremented vectors $\boldsymbol{\phi}_{i+1}$ and $\mathbf{t}_{i+1}$, the shell governing equations of motion are satisfied at the (i+1)-th step as

$$\bar{\rho}\,\ddot{\boldsymbol{\phi}}_{i+1} + \bar{c}\,\dot{\boldsymbol{\phi}}_{i+1} = -(\mathbf{R}_f)_{i+1} + \bar{\mathbf{n}}_{i+1}, \tag{33}$$

$$\bar{\mathbf{I}}_p\,\ddot{\boldsymbol{\theta}}_{i+1} + \bar{\mathbf{I}}_c\,\dot{\boldsymbol{\theta}}_{i+1} = -(\mathbf{R}_m)_{i+1} + (\mathbf{m}_e)_{i+1}. \tag{34}$$

Subtracting Eqs. (32) and (33) from Eqs. (27) and (28), respectively, we obtain

$$\bar{\rho}\,\Delta\ddot{\boldsymbol{\phi}}_i + \bar{c}\,\Delta\dot{\boldsymbol{\phi}}_i = -(\Delta\mathbf{R}_f)_i + \Delta\bar{\mathbf{n}}_i, \tag{35}$$

$$\bar{\mathbf{I}}_p\,\Delta\ddot{\boldsymbol{\theta}}_i + \bar{\mathbf{I}}_c\,\Delta\dot{\boldsymbol{\theta}}_i = -(\Delta\mathbf{R}_m)_i + (\Delta\mathbf{m}_e)_i. \tag{36}$$

The changes in the variables may be expressed as follows: $\Delta\boldsymbol{\phi}_i = \boldsymbol{\phi}_{i+1} - \boldsymbol{\phi}_i$, $\Delta\ddot{\boldsymbol{\theta}}_i = \ddot{\boldsymbol{\theta}}_{i+1} - \ddot{\boldsymbol{\theta}}_i$, $(\Delta\mathbf{R}_f)_i = (\mathbf{R}_f)_{i+1} - (\mathbf{R}_f)_i$, $(\Delta\mathbf{R}_m)_i = (\mathbf{R}_m)_{i+1} - (\mathbf{R}_m)_i$, $\Delta\bar{\mathbf{n}}_i = \bar{\mathbf{n}}_{i+1} - \bar{\mathbf{n}}_i$, and $(\Delta\mathbf{m}_e)_i = (\mathbf{m}_e)_{i+1} - (\mathbf{m}_e)_i$.

Using the Newmark-beta method, $\Delta\dot{\boldsymbol{\phi}}_i$, $\Delta\ddot{\boldsymbol{\phi}}_i$, $\Delta\dot{\boldsymbol{\theta}}_i$, and $\Delta\ddot{\boldsymbol{\theta}}_i$ can be written as:

$$\Delta\dot{\boldsymbol{\phi}}_i = \frac{\gamma}{\beta\Delta t}\Delta\boldsymbol{\phi}_i - \frac{\gamma}{\beta}\dot{\boldsymbol{\phi}}_i + \Delta t\left(1 - \frac{\gamma}{2\beta}\right)\ddot{\boldsymbol{\phi}}_i, \tag{37}$$

$$\Delta\ddot{\boldsymbol{\phi}}_i = \frac{1}{\beta\Delta t^2}\Delta\boldsymbol{\phi}_i - \frac{1}{\beta\Delta t}\dot{\boldsymbol{\phi}}_i - \frac{1}{2\beta}\ddot{\boldsymbol{\phi}}_i, \tag{38}$$

$$\Delta\dot{\boldsymbol{\theta}}_i = \frac{\gamma}{\beta\Delta t}\Delta\boldsymbol{\theta}_i - \frac{\gamma}{\beta}\dot{\boldsymbol{\theta}}_i + \Delta t\left(1 - \frac{\gamma}{2\beta}\right)\ddot{\boldsymbol{\theta}}_i, \tag{39}$$

$$\Delta\ddot{\boldsymbol{\theta}}_i = \frac{1}{\beta\Delta t^2}\Delta\boldsymbol{\theta}_i - \frac{1}{\beta\Delta t}\dot{\boldsymbol{\theta}}_i - \frac{1}{2\beta}\ddot{\boldsymbol{\theta}}_i. \tag{40}$$

In this work, we used the Newmark-beta method to solve the PD governing equations of motion. There are two parameters in this method, viz., β and γ, that need to be specified. This method is implicit if $\beta \neq 0$. For a linear dynamical system, it is unconditionally stable if the following criterion is satisfied: $2\beta \geq \gamma \geq 1/2$ [39]. However, for a nonlinear dynamical system, as in our case, there is no such criterion to ensure stability. In the present work, we used $\beta = 1/6$ and $\gamma = 1/2$, which makes this method the same as the widely used linear acceleration method [40]. The implicit nature of this method helps in choosing higher time steps compared to explicit time-marching schemes. This is particularly important in our case as we used polymeric materials in which wave propagation speed is low and we desire to see the wave propagation for a longer period of time.

Using Eqs. (36) through (39) in Eqs. (34) and (35), one obtains:

$$\left(\frac{6\bar{\rho}}{\Delta t^2} + \frac{3\bar{c}}{\Delta t}\right)\Delta\boldsymbol{\phi}_i = \Delta\bar{\mathbf{n}}_i - (\Delta\mathbf{R}_f)_i + \left(\frac{6\bar{\rho}}{\Delta t} + 3\bar{c}\right)\dot{\boldsymbol{\phi}}_i + \left(3\bar{\rho} + \frac{\bar{c}\Delta t}{2}\right)\ddot{\boldsymbol{\phi}}_i, \tag{41}$$

$$\left(\frac{6\bar{\mathbf{I}}_p}{\Delta t^2} + \frac{3\bar{\mathbf{I}}_c}{\Delta t}\right)\Delta\boldsymbol{\theta}_i = (\Delta\mathbf{m}_e)_i - (\Delta\mathbf{R}_m)_i + \left(\frac{6\bar{\mathbf{I}}_p}{\Delta t} + 3\bar{\mathbf{I}}_c\right)\dot{\boldsymbol{\theta}}_i + \left(3\bar{\mathbf{I}}_p + \frac{\bar{\mathbf{I}}_c\Delta t}{2}\right)\ddot{\boldsymbol{\theta}}_i. \tag{42}$$

$\Delta\mathbf{R}_f$ and $\Delta\mathbf{R}_m$ depend nonlinearly on $\Delta\boldsymbol{\phi}$ and $\Delta\theta$. In order to solve Eqs. (40) and (41), the Newton-Raphson method is used. Assuming at the n-th iteration, these equations are satisfied, guess values of $\Delta\boldsymbol{\phi}_i^n$ and $\Delta\theta_i^n$ are chosen to proceed to the $(n+1)$-th iteration. As Eqs. (40) and (41) do not satisfy with $\Delta\boldsymbol{\phi}_i^n$ and $\Delta\theta_i^n$ values, corrections are applied to them as:

$$\Delta\boldsymbol{\phi}_i^{n+1} = \Delta\boldsymbol{\phi}_i^n + \delta\Delta\boldsymbol{\phi}_i, \tag{43}$$

$$\Delta\theta_i^{n+1} = \Delta\theta_i^n + \delta\Delta\theta_i. \tag{44}$$

As $\Delta\boldsymbol{\phi}_i^{n+1}$ and $\Delta\theta_i^{n+1}$ exactly satisfy Eqs. (40) and (41), we obtain

$$\left(\frac{6\bar{\rho}}{\Delta t^2}+\frac{3\bar{c}}{\Delta t}\right)\Delta\boldsymbol{\phi}_i^{n+1} = \Delta\bar{\mathbf{n}}_i - (\Delta\mathbf{R}_f)_i^{n+1} + \left(\frac{6\bar{\rho}}{\Delta t}+3\bar{c}\right)\dot{\boldsymbol{\phi}}_i + \left(3\bar{\rho}+\frac{\bar{c}\Delta t}{2}\right)\ddot{\boldsymbol{\phi}}_i, \tag{45}$$

$$\left(\frac{6\bar{\mathbf{I}}_p}{\Delta t^2}+\frac{3\bar{\mathbf{I}}_c}{\Delta t}\right)\Delta\theta_i^{n+1} = (\Delta\mathbf{m}_e)_i - (\Delta\mathbf{R}_m)_i^{n+1} + \left(\frac{6\bar{\mathbf{I}}_p}{\Delta t}+3\bar{\mathbf{I}}_c\right)\dot{\theta}_i + \left(3\bar{\mathbf{I}}_p+\frac{\bar{\mathbf{I}}_c\Delta t}{2}\right)\ddot{\theta}_i, \tag{46}$$

where $(\Delta\mathbf{R}_f)_i^{n+1}$ and $(\Delta\mathbf{R}_m)_i^{n+1}$ can be written as

$$(\Delta\mathbf{R}_f)_i^{n+1} = \mathbf{R}_f\left(\boldsymbol{\phi}_i+\Delta\boldsymbol{\phi}_i^n+\delta\Delta\boldsymbol{\phi}_i, \exp\left(\text{skew}(\delta\Delta\theta_i)\exp\left(\text{skew}(\Delta\theta_i^n)\right)\mathbf{t}_i\right)\right) - \mathbf{R}_f(\boldsymbol{\phi}_i,\mathbf{t}_i), \tag{47}$$

$$(\Delta\mathbf{R}_m)_i^{n+1} = \mathbf{R}_m\left(\boldsymbol{\phi}_i+\Delta\boldsymbol{\phi}_i^n+\delta\Delta\boldsymbol{\phi}_i, \exp\left(\text{skew}(\delta\Delta\theta_i)\exp\left(\text{skew}(\Delta\theta_i^n)\right)\mathbf{t}_i\right)\right) - \mathbf{R}_m(\boldsymbol{\phi}_i,\mathbf{t}_i). \tag{48}$$

Expanding $(\Delta\mathbf{R}_f)_i^{n+1}$ and $(\Delta\mathbf{R}_m)_i^{n+1}$ in Taylor series, keeping the first-order term, and using the relation $\theta = \Lambda\Theta$, we finally arrive at the linearized version of Eqs. (44) and (45) as:

$$\left[\left(\frac{6\bar{\rho}}{\Delta t^2}+\frac{3\bar{c}}{\Delta t}\right)\mathbf{I} + \left.\frac{\partial\mathbf{R}_f}{\partial\boldsymbol{\phi}}\right|_{\left(\boldsymbol{\phi}_i+\Delta\boldsymbol{\phi}_i^n,\exp\left(\text{skew}(\Delta\theta_i^n)\mathbf{t}_i\right)\right)}\right]\delta\Delta\boldsymbol{\phi}_i + \left.\frac{\partial\mathbf{R}_f}{\partial\theta}\right|_{\left(\boldsymbol{\phi}_i+\Delta\boldsymbol{\phi}_i^n,\exp\left(\text{skew}(\Delta\theta_i^n)\mathbf{t}_i\right)\right)}\Lambda_i\delta\Delta\Theta_i$$
$$= \Delta\bar{\mathbf{n}}_i - (\Delta\mathbf{R}_f)_i^n + \left(\frac{6\bar{\rho}}{\Delta t}+3\bar{c}\right)\dot{\boldsymbol{\phi}}_i + \left(3\bar{\rho}+\frac{\bar{c}\Delta t}{2}\right)\ddot{\boldsymbol{\phi}}_i - \left(\frac{6\bar{\rho}}{\Delta t^2}+\frac{3\bar{c}}{\Delta t}\right)\Delta\boldsymbol{\phi}_i^n, \tag{49}$$

$$\Lambda^\top\left.\frac{\partial\mathbf{R}_m}{\partial\boldsymbol{\phi}}\right|_{\left(\boldsymbol{\phi}_i+\Delta\boldsymbol{\phi}_i^n,\exp\left(\text{skew}(\Delta\theta_i^n)\mathbf{t}_i\right)\right)}\delta\Delta\boldsymbol{\phi}_i + \Lambda^\top\left[\left(\frac{6\bar{\mathbf{I}}_p}{\Delta t^2}+\frac{3\bar{\mathbf{I}}_c}{\Delta t}\right)\mathbf{I} + \left.\frac{\partial\mathbf{R}_m}{\partial\theta}\right|_{\left(\boldsymbol{\phi}_i+\Delta\boldsymbol{\phi}_i^n,\exp\left(\text{skew}(\Delta\theta_i^n)\mathbf{t}_i\right)\right)}\right]\Lambda_i\delta\Delta\Theta_i$$
$$= \Lambda^\top\left[(\Delta\mathbf{m}_e)_i - (\Delta\mathbf{R}_m)_i^n + \left(\frac{6\bar{\mathbf{I}}_p}{\Delta t}+3\bar{\mathbf{I}}_c\right)\Lambda\dot{\Theta}_i + \left(3\bar{\mathbf{I}}_p+\frac{\bar{\mathbf{I}}_c\Delta t}{2}\right)\Lambda\ddot{\Theta}_i - \left(\frac{6\bar{\mathbf{I}}_p}{\Delta t^2}+\frac{3\bar{\mathbf{I}}_c}{\Delta t}\right)\Lambda\Delta\Theta_i^n\right]. \tag{50}$$

By writing Eqs. (48) and (49) for all the particles in the PD domain, $\delta\Delta\boldsymbol{\phi}_i$ and $\delta\Delta\Theta_i$ can be solved. Newton–Raphson iterations should be performed till $\Delta\boldsymbol{\phi}_i^{n+1}$ and $\Delta\theta_i^{n+1}$ in Eqs. (42) and (43) satisfy Eqs. (44) and (45) within a tolerance limit.

5. Experimental methods

5.1. 3D printing and parameter selection

Metamaterial structures are modeled using SolidWorks 2025 and subsequently fabricated using a 3D printer (Make & Model: Ultimaker 3 Extended) that works on a fused deposition method (FDM) and has a printing accuracy of $\pm$ 0.1 mm. In this process, the thermoplastic filament is heated to its semi-molten state and extruded through a nozzle, depositing material in a predetermined path to form each layer, which cumulatively builds the final structure. The opted 3D printing parameters included a layer thickness of 0.1 mm and a 100% infill density of the pattern. FDM is known to induce porosity even after considering the maximum infill density (i.e. 100%), therefore, to reduce it as much as possible, wall line count of 100 along with wall line thickness of 0.1mm has been considered. Moreover, it is observed that the mechanical properties of the printed material are affected by factors such as layer orientation, infill density, and infill pattern [37, 38]. To maintain consistency, all sample structures shown in Fig. 3 were printed using the same specifications listed in Table 1. The thermoplastic polyurethane (TPU) filament used for specimen fabrication had a diameter of 2.85 mm. The material possesses a density of 1200 $\mathrm{kg/m^3}$ and a Poisson's ratio of 0.4.

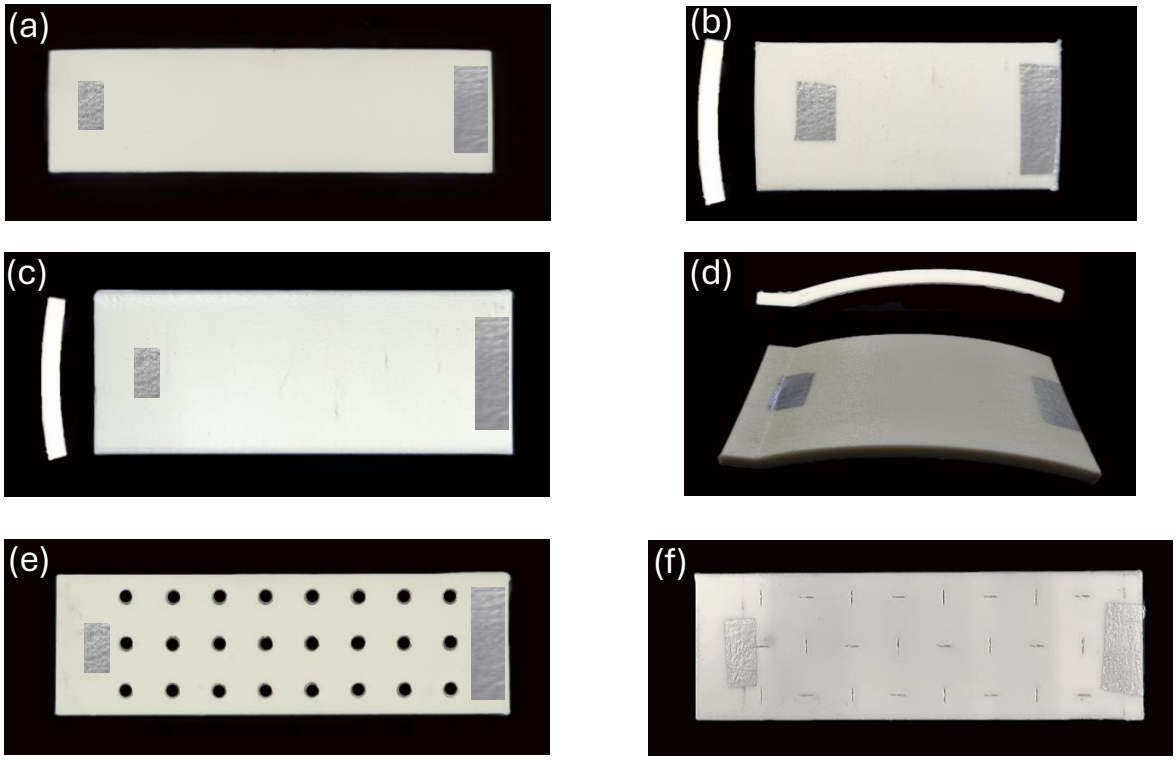


Fig. 3: TPU 95A 3D-printed plate and shell structures: (a) solid plate, (b) solid shell configuration 1, (c) solid shell configuration 2, and (d) solid shell configuration 3. (e) plate with periodic circular holes (presented in 3×8 pattern), and (f) plate with periodic horizontal and vertical cracks.

5.2. Experimental results to determine material properties

The Young's Modulus is a fundamental mechanical property that measures the stiffness of a material or its resistance to elastic deformation. To determine the Young's Modulus of the 3D-printed TPU 95A material, a standardized tensile test is performed using a Universal Testing Machine (UTM). A sample is prepared in the shape of a dogbone, following the specifications

Table 1: A detailed specification of 3D printing process while printing samples.

Specification in 3D printing	Values
3D printing material	TPU 95A
Layer thickness	0.1 mm
Infill pattern	Grid
Infill density	100%
Printing speed	15 mm/sec
Print temperature	225°C
Build plate temperature	0°C
Wall line thickness	0.1 mm
Wall line count	100
Top and bottom layer thickness	0.4 mm

of a recognized ASTM standard (ASTM D638 for polymers). The sample is stretched at a constant rate of 2 mm/min, and the resultant stress-strain curve is plotted. This slow, quasi-static loading rate ensures that the material's inherent properties are measured, as it minimizes any time-dependent or dynamic effects. The Young's Modulus determined from the stress-strain curve is 19 MPa.

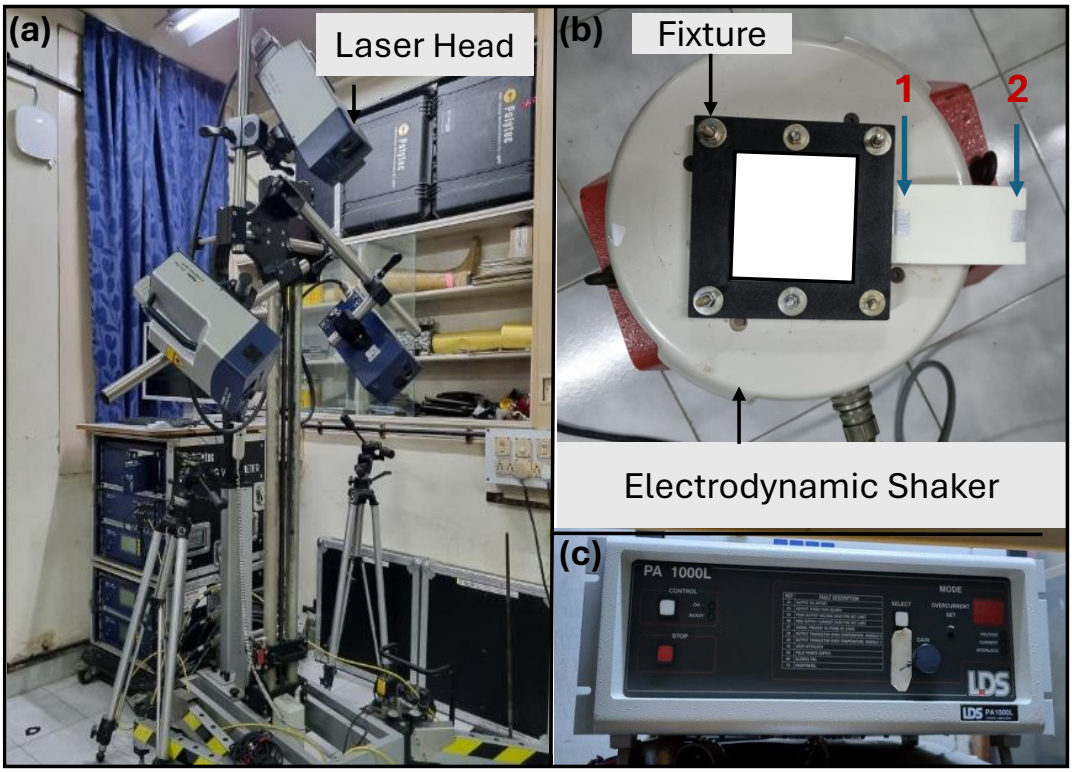


Fig. 4: Experimental setup: (a) Laser Doppler Vibrometer (LDV) with three scanning heads (Polytec PSV400), (b) an electrodynamic shaker (LDS V455) with a solid shell mounted on it via a fixture, and (c) power amplifier.

5.3. Experimental procedure to determine displacement transmissibility

To validate the proposed model, experiments were conducted using a Laser Doppler Vibrometer (LDV, PSV400) to measure the response of solid plates excited by an electrodynamic

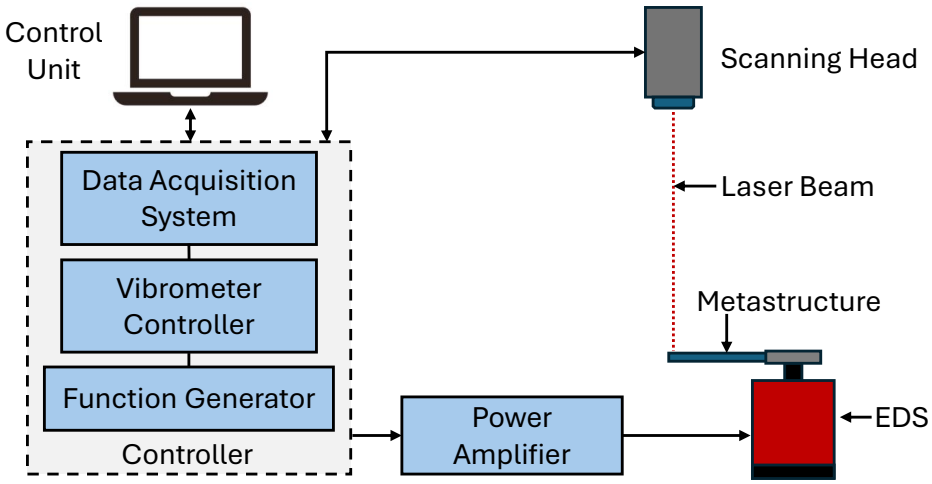


Fig. 5: Schematic diagram of the experimental setup: a laser head scans the signal at the defined points on the plate/shell surface, while the electrodynamic shaker, driven by a power amplifier, applies the desired vibration to the shell.

shaker (LDS V455). The shaker provides sinusoidal excitation at the defined frequencies, with a gain and amplification factor of 1, utilizing the sampling rate of 4000 data points per second. The experimental setup and corresponding schematic diagram are presented in Fig. 4 and Fig. 5, respectively. The plate was mounted on the shaker via a fixture that provides the base excitation. A 10 mm extension was added to the structures to facilitate clamping with the fixture. The curved edge plate accurately mimics the shape of the fixture. During the experiments, response measurements for the incident and transmitted signals were taken at two points on the mechanical metasurface: one near the base and another at the free end. Both points were positioned 5 mm from the edge. To minimize laser scattering, retroreflective tape was applied at both measuring locations [34].

5.4. Damping calculation by half-power bandwidth method

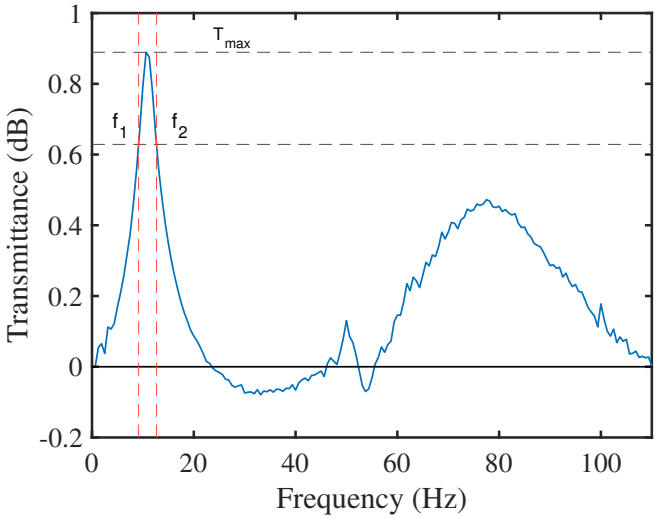


Fig. 6: Determination of damping of TPU using half-power bandwidth method.

Damping holds significant importance in mechanical systems for accurate modeling of energy dissipation. In half-power bandwidth method, the structure's base is excited using electro-

dynamic shaker and the response signals are recorded in the frequency domain using an LDV. First, the resonant peak is identified, and then the bandwidth corresponding to the amplitude at $1/\sqrt{2}$ of the resonant peak is determined, as shown in Fig. 6. The damping ratio is calculated by the following formula:

$$2\zeta = \frac{f_2 - f_1}{f_n}, \tag{51}$$

where f_n is the natural resonant frequency, and f_1 and f_2 are the frequencies corresponding to $1/\sqrt{2}$ of resonant peak amplitude. The damping ratio (ζ) evaluated from the test using the half-power bandwidth method is found to be 0.16.

6. Numerical simulations

Dynamic response of plates and shells are investigated using the proposed Simo-Reissner PD shell theory with damping. Time integration of the governing equations is carried out using the Newmark-beta method, and convergence at each time step is ensured through the Newton-Raphson method. The plates and shells are spatially discretized into particles, and boundary conditions are imposed on a single layer of particles. Accuracy of the proposed model is assessed by comparing PD solutions with FEM results obtained from ABAQUS® using the S4R shell element. The PD horizon size is defined as $\delta = 1.01\sqrt{r^2\Delta\theta^2 + \Delta z^2}$, where r is the shell radius, and $\Delta\theta$ and Δz are the azimuthal and axial grid spacing, respectively.

The simulations are carried out using a workstation with two CPUs-Intel Xeon 16 Core Gold Series 6226R processor with speed 2.9 GHz and 128 GB RAM. The computer program is written in FORTRAN 90 and compiled using Intel® Fortran Compiler. The graphs are plotted using MATLAB®, schematic diagrams are prepared using INKSCAPE, and ParaView is used to plot the contours. Typically, one PD simulation takes approximately 4 days to complete for the problems shown in this paper. One FE simulation takes approximately 12 hours to complete using parallel processing in ABAQUS® in this work. In the case of the PD models, we do not employ parallel processing, which results in larger computation time.

6.1. Solid plates

Two TPU 95A plate configurations are considered, viz., one with a straight edge and the other with a curved edge where the excitation is applied. The straight-edge plate has a length of 115 mm, width of 40 mm, thickness of 4 mm, and radius of curvature of 1000 mm as shown in Fig. 7(c). Note that plates are modeled as shells with a large radius of curvature. The curved-edge plate has the same dimensions as the straight-edge plate except that its one of the edges is curved as shown in Fig. 7(d). Material properties of TPU 95A are as follows: Young's modulus $E = 19$ MPa, Poisson's ratio $\nu = 0.4$, density $\rho = 1200\,\mathrm{kg/m^3}$, and Rayleigh damping coefficient $\alpha = 409.589\ \mathrm{s^{-1}}$. The plates are subjected to displacement-controlled transverse excitation at one edge with excitation frequencies of 200 Hz and 800 Hz, keeping all the other degrees of freedom fixed. Apart from the loading edge, all the other edges are free as illustrated in Figs. 7(a) and 7(b).

At 200 Hz excitation, displacement time histories are compared for PD and FEM with damping at point O_1 [Fig. 7], and experiments for both straight-edge and curved-edge plates as shown in Figs. 8(a) through 8(d). For the straight-edge plate, the absolute peak displacement values in the experiment, FEM, and PD simulations are 2.5837×10^{-5} m, 2.0209×10^{-5} m, and 1.9328×10^{-5} m, respectively. Similarly, for the curved-edge plate, the corresponding values are 2.5837×10^{-5} m, 2.1510×10^{-5} m, and 1.9311×10^{-5} m. In both cases, the PD and FEM

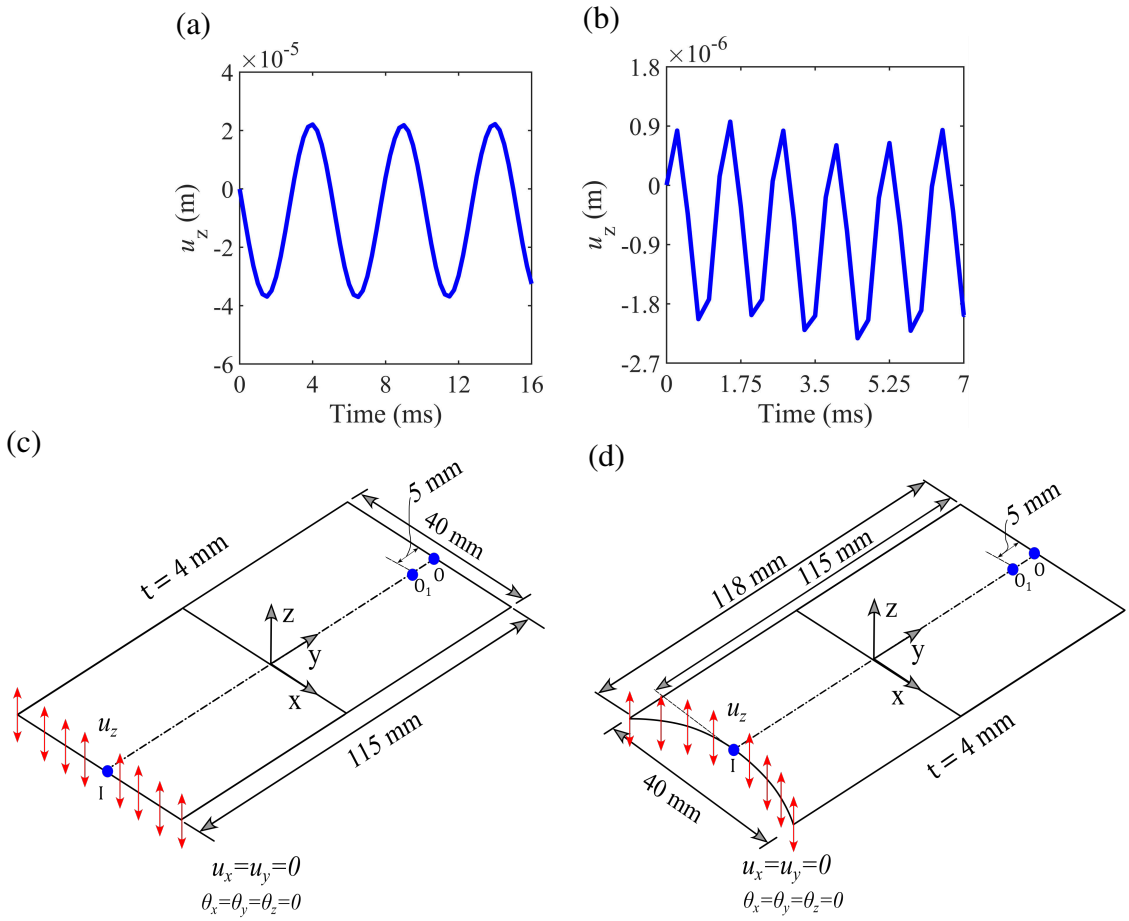


Fig. 7: The experimental input transverse displacement time histories for different excitation frequencies: at (a) 200 Hz, and (b) 800 Hz. Geometry, boundary and loading conditions applied for TPU 95A plates: (c) rectangular plate with a straight edge, and (d) rectangular plate with a curved edge.

results are in good agreement and closely match with the experimental data, demonstrating the fidelity of the proposed PD shell theory. The phase portraits compare the experimental input, experimental output, PD output, and FEM output for excitation frequencies of 200 Hz for both straight-edge and curved-edge plates, as presented in Figs. 8(e) and 8(f).

Convergence analysis is performed for the PD model by carrying out simulations on the straight-edge plate excited at 800 Hz. Three types of discretization are considered with 2697, 7293, and 23569 material points, and they are denoted as PD30, PD50, and PD90, respectively, where the 30, 50, and 90 represent number of divisions in the width direction. Transverse displacement time histories at the output point O_1 are compared against the converged FEM solution with 23220 elements as presented in Fig. 9, which shows a good agreement.

At 800 Hz excitation, the transverse displacement response is analyzed in both time and frequency domains. For the straight-edge plate, PD results are compared with FEM in the presence of damping at point O_1 [Fig. 7(a)] and shown in Figs. 11(a) and 11(b). For the curved-edge plate, PD results with damping are compared with FEM and experimental measurements at point O_1 [Fig. 7(b)] as depicted in Figs. 11(c) and 11(d). The absolute peak displacement values in the experiment, FEM, and PD simulations are 6.9463×10^{-7} m, 3.8793×10^{-7} m, and 4.5815×10^{-7} m. The phase portraits illustrate the comparison between the experimental input, experimental output, PD output, and FEM output at excitation frequencies of 800 Hz for both straight-edge and curved-edge plates, corresponding to Figs. 11(e) and 11(f), respectively. The

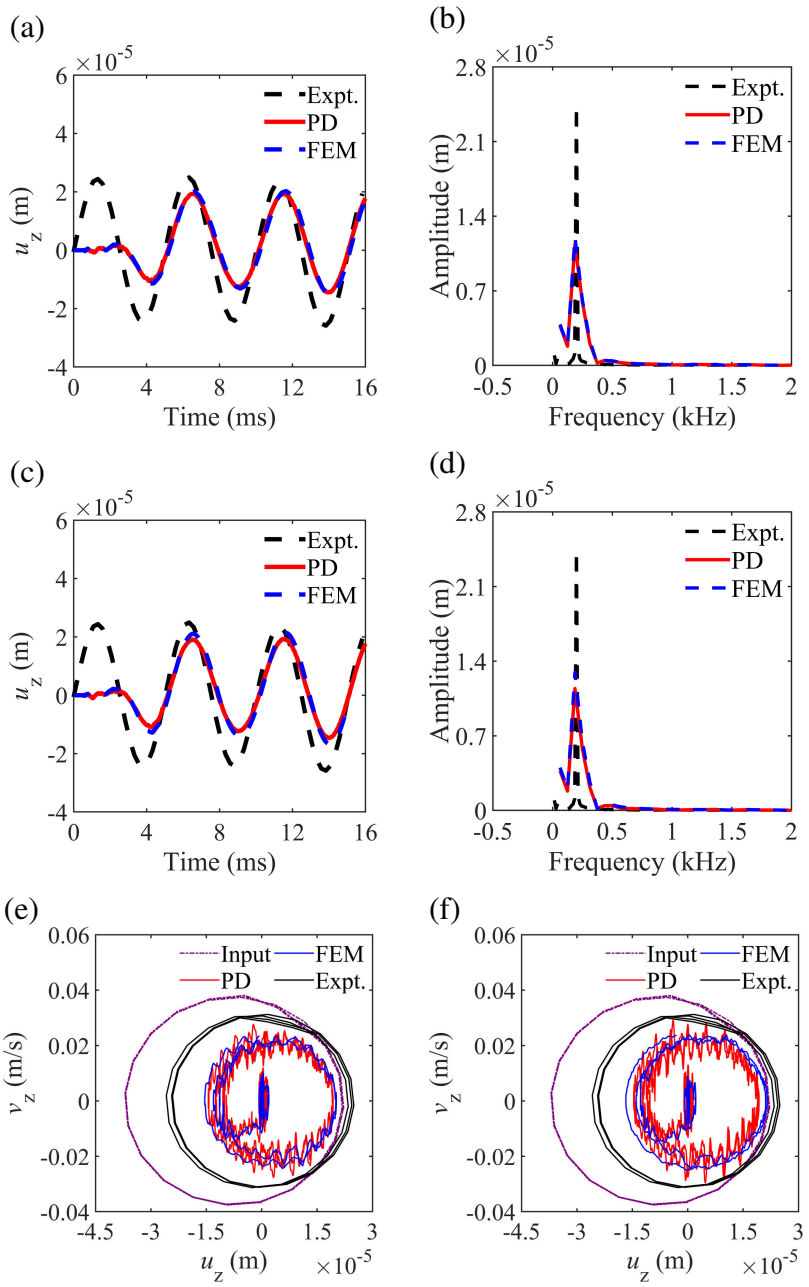


Fig. 8: Transverse displacement response (m) of TPU solid plates under 200 Hz excitation with damping. Comparison of experiment, PD, and FEM results: straight-edge plate: (a) time history and (b) amplitude spectrum; curved-edge plate: (c) time history and (d) amplitude spectrum. Phase portraits: (e) straight-edge plate and (f) curved-edge plate.

PD and FEM results agree well and closely match the experimental data.

Transverse displacement contours of the curved-edge plate under 800 Hz excitation at various times are presented in Figs. 10(a) through 10(d) which illustrate wave propagation over time, showing progression of the wavefront and interaction of wave with the free boundary.

6.2. Solid shells

This section comprises three different configurations of solid shell. TPU 95A material is used for all the configurations, considering damping. These shells are subjected to displacement-controlled loading with excitation frequencies of 100 Hz, 200 Hz, and 800 Hz at one edge in

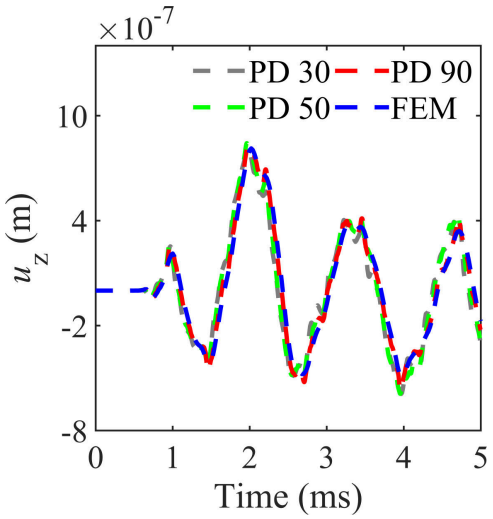


Fig. 9: Convergence study for the PD model considering various discretization (PD30, PD50, and PD90) and comparison with FEM for 800 Hz excitation.

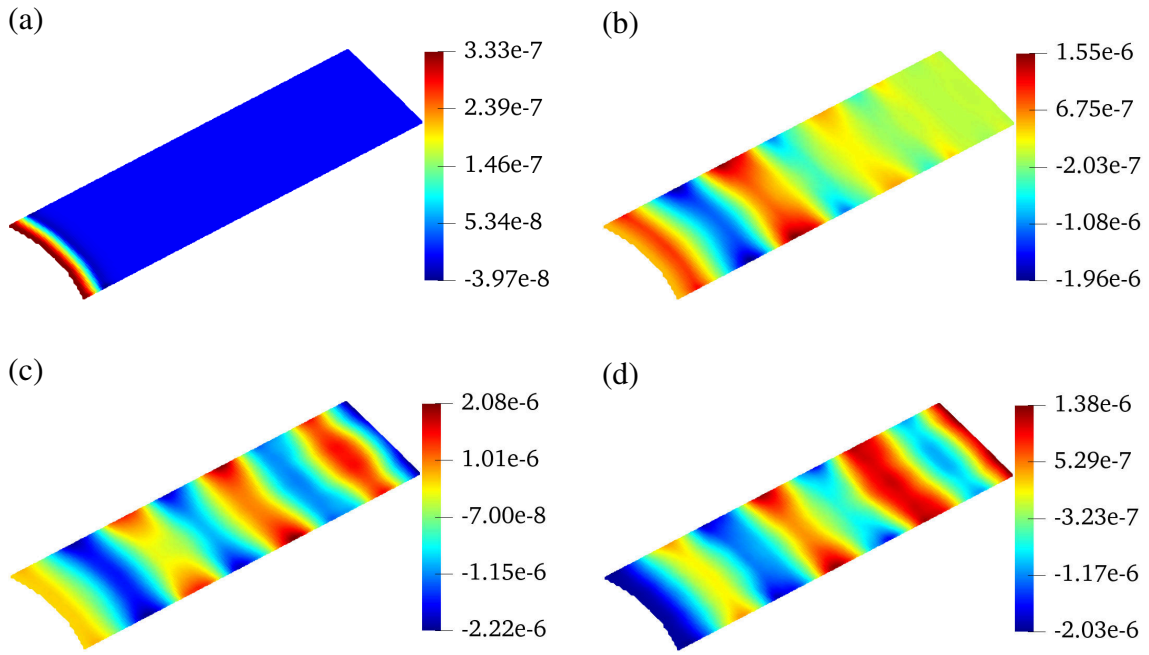


Fig. 10: Transverse displacement (u_z) contours (m) for TPU curved-edge plate under 800 Hz excitation, showing wave propagation at successive times: (a) t = 0.1 ms, (b) t = 1.6 ms, (c) t = 4 ms, and (d) t = 7 ms.

the transverse direction, keeping all other degrees of freedom fixed. The remaining edges of the shell are free.

Solid shell configuration 1 has length of 45 mm, span of 35 mm, thickness of 4 mm, radius of curvature of 100 mm, and Rayleigh damping coefficient of $\alpha = 894.795$ s^{-1}. The shell is subjected to excitation frequencies of 200 Hz and 800 Hz, as shown in Figs. 12(a) and 12(b), respectively. Displacements are applied at the input point (I) in the midpoint of the loading edge, and measured at the output point (O_1), which lies along the y-axis at a distance of 5 mm from the free edge of the solid shell configuration 1, as indicated in Figs. 12(c). The PD model, consisting of 7979 material points, and the FEM model, comprising 8160 elements, are analyzed under displacement-controlled transverse loading.

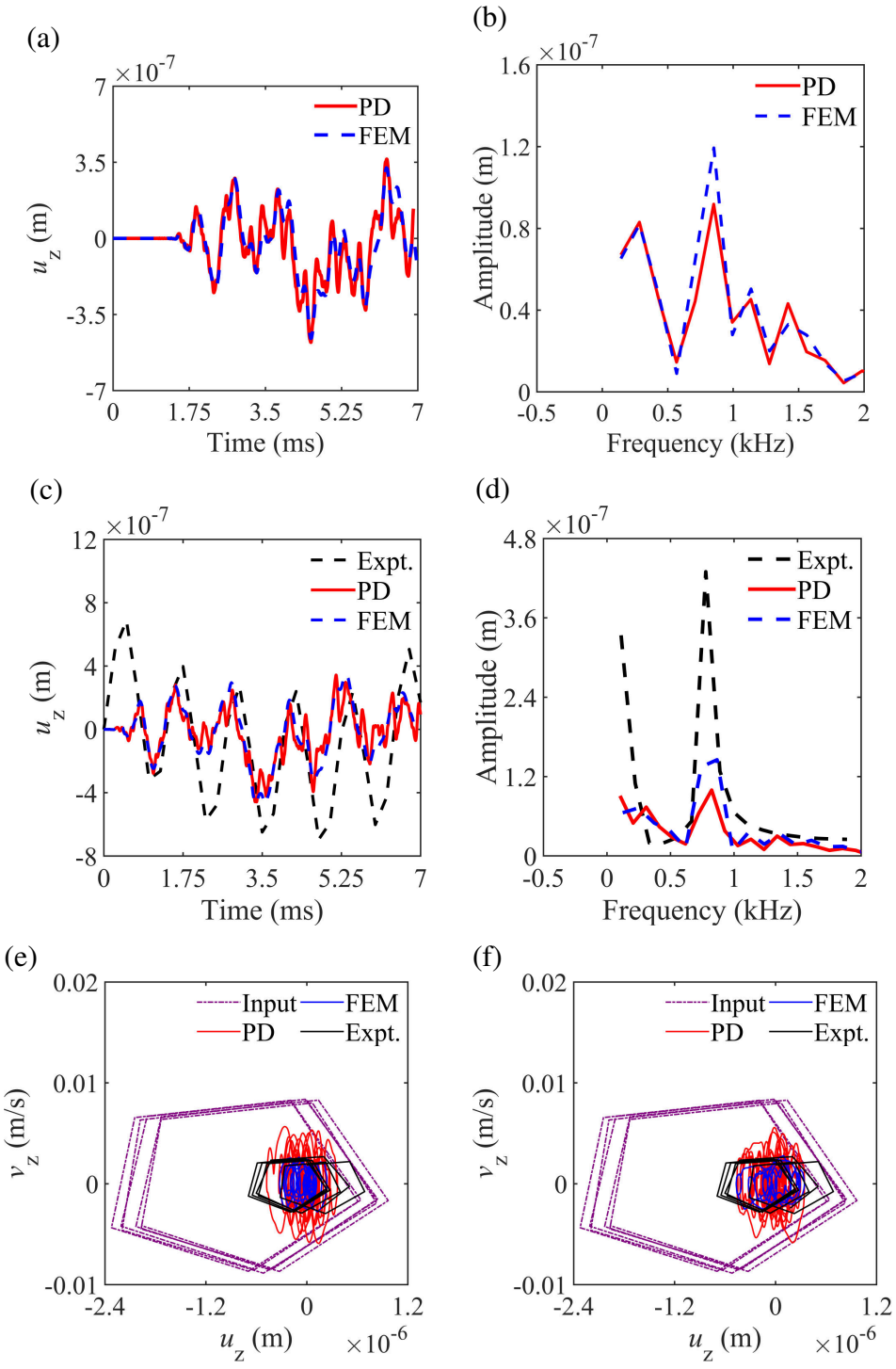


Fig. 11: Transverse displacement response (m) of solid TPU plates subjected to displacement-controlled loading of 800 Hz frequency; straight-edge plate with damping: (a) time history and (b) amplitude spectrum; curved-edge plate with damping: (c) time history and (d) amplitude spectrum. Phase portraits of (e) straight-edge plate and (f) curved-edge plate.

The displacement time histories obtained from the PD and FEM models, both incorporating damping, are compared with experimental measurements at location O_1 [Fig. 12] for excitation frequencies of 200 Hz and 800 Hz, as shown in Figs. 13(a) through 13(d). For the 200 Hz excitation, the maximum displacement values from the experiment, FEM, and PD simulations

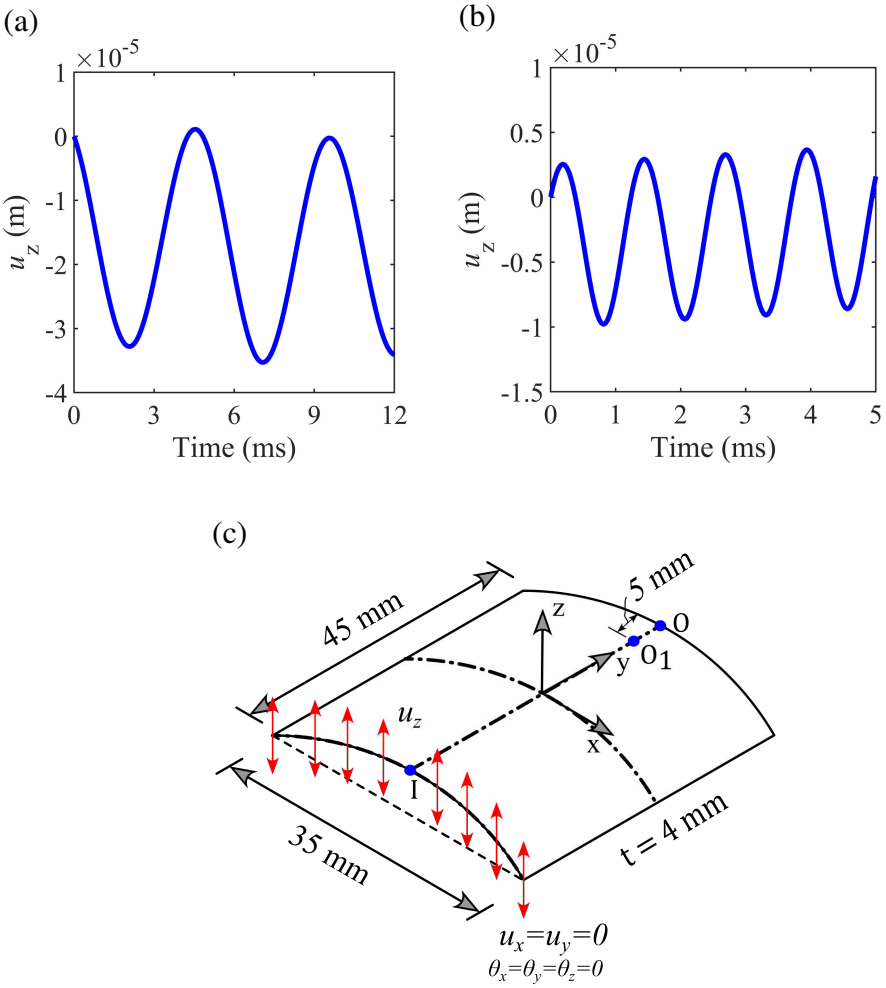


Fig. 12: Experimental transverse displacement inputs used in the simulation of the shell configuration 1 for excitation frequencies of (a) 200 Hz, and (b) 800 Hz. (c) Reference configuration and boundary conditions of the solid shell configuration 1.

are given as 2.0301×10^{-5} m, 1.8482×10^{-5} m, and 1.9212×10^{-5} m, respectively. Similarly, for the 800 Hz excitation, the corresponding peak values are 5.9312×10^{-6} m, 3.4245×10^{-6} m, and 3.3163×10^{-6} m. A comparison of the experimental input, experimental output, PD output, and FEM output under excitation frequencies of 200 Hz and 800 Hz is presented in the phase portraits shown in Figs. 13(e) and 13(f), respectively. In both cases, the PD and FEM results show good agreement with the experimental data, thereby validating the accuracy of the proposed PD shell theory.

Transverse displacement contours of the solid shell configuration 1 subjected to 200 Hz excitation at different time instants are shown in Figs. 14(a) through 14(d). These results illustrate the wave propagation, highlighting the progression of the wavefront and its interaction with the free boundary.

Solid shell configuration 2 is examined next. It has an overall length of 80 mm, span of 35 mm, uniform thickness of 4 mm, radius of curvature of 100 mm, and Rayleigh damping coefficient of $\alpha = 503.322$ s^{-1}. The structure is excited at frequencies of 100 Hz and 200 Hz, as illustrated in Figs. 15(a) and 15(b), respectively. The transverse response is measured at two locations: the input point (I), positioned at the center of the loaded edge, and the output point (O_1), located along the y-axis at a distance of 5 mm from the free edge, as shown in Fig. 15(c). Numerical simulations are performed using both PD with 14299 material points

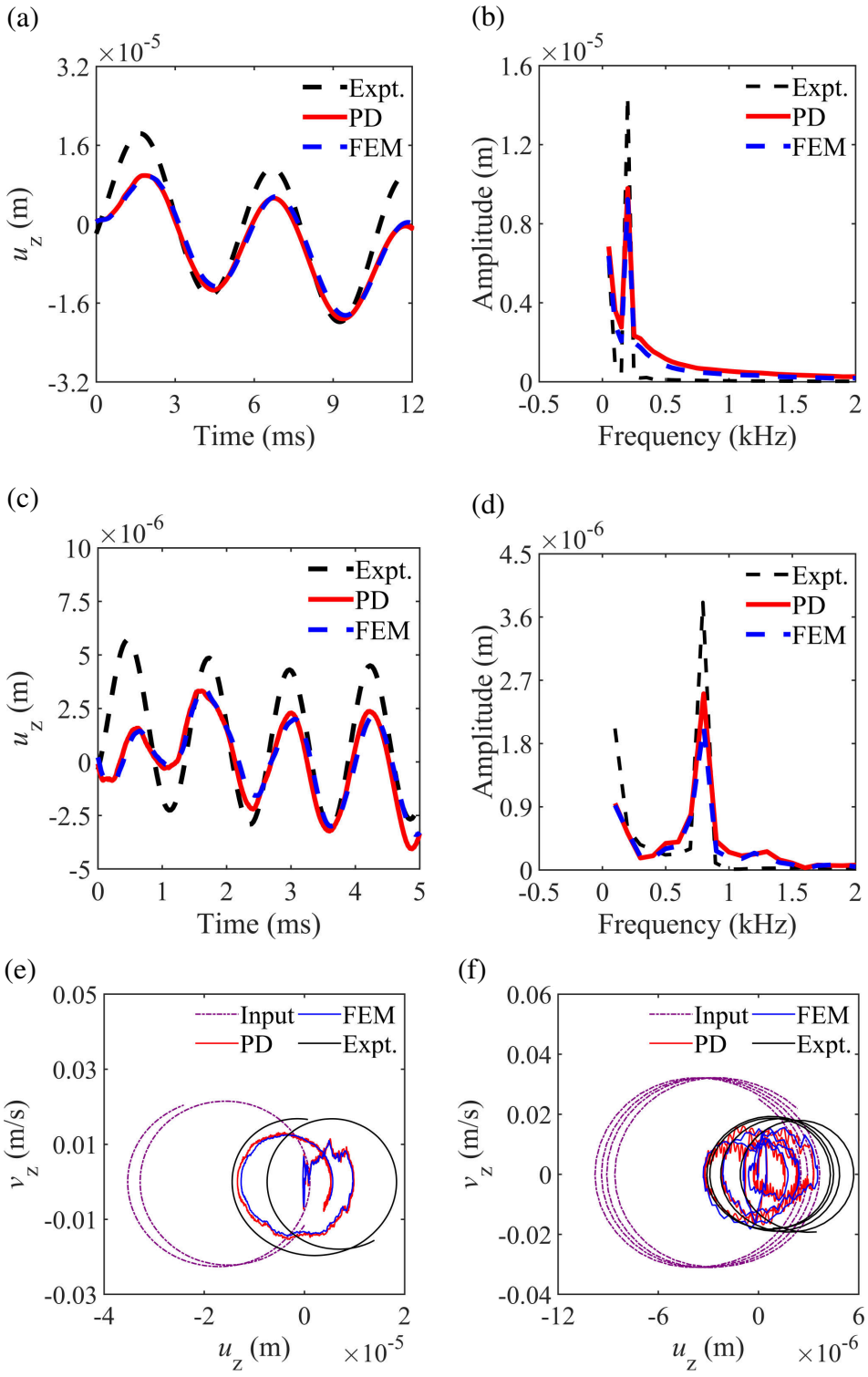


Fig. 13: Transverse displacement response (m) of TPU solid shell configuration 1 with damping. Comparison of experimental, PD, and FEM results: for 200 Hz excitation, (a) time history and (b) amplitude spectrum; for 800 Hz excitation, (c) time history and (d) amplitude spectrum. Phase portraits of transverse response under excitation frequencies of (e) 200 Hz and (f) 800 Hz.

and FEM consisting of 14560 elements, each subjected to displacement-controlled transverse excitation.

The displacement time histories for both the PD and FEM models, including damping,

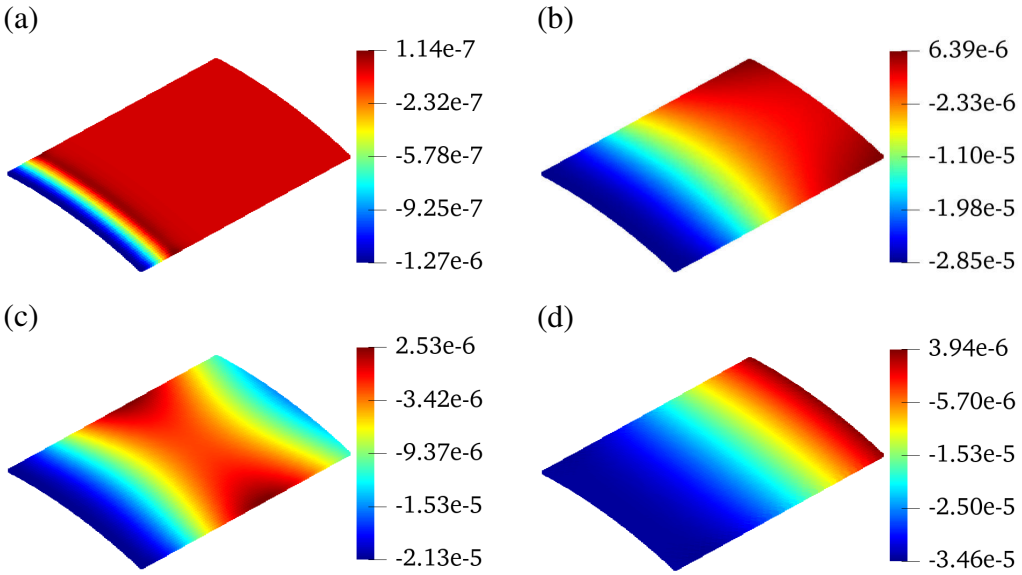


Fig. 14: Transverse displacement u_z contours (m) for a TPU solid shell configuration 1. Under 200 Hz excitation: (a) t = 0.1 ms, (b) t = 1.5 ms, (c) t = 6 ms, and (d) t = 12 ms.

are compared at location O_1[Fig. 15], alongside experimental data for both 100 Hz and 200 Hz excitation frequencies, as shown in Figs. 16(a) through 16(d). For the 100 Hz excitation, the maximum displacement values observed in the experiments, FEM, and PD simulations are 3.0417×10^{-5} m, 2.2759×10^{-5} m, and 2.2346×10^{-5} m, respectively. Similarly, for the 200 Hz excitation, the corresponding peak values are noted as 1.9698×10^{-5} m, 7.6952×10^{-6} m, and 7.4617×10^{-6} m. Figs. 16(e) and 16(f) present phase portraits comparing the experimental input, experimental output, PD output, and FEM output under excitation frequencies of 100 Hz and 200 Hz, respectively. In both cases, the results from the PD and FEM simulations are close to the experimental data, verifying the accuracy of the proposed PD shell theory. The discrepancy between the simulation and experimental results may be attributed to several factors, including small rigid-body motion of the plate during the experiment, dynamic resonance effects that may induce localized instability in certain regions of the plate, measurement uncertainties, inaccuracies in the estimation of material properties (such as Young's modulus, Poisson's ratio, and damping ratio), and manufacturing-related imperfections or constraints. We performed the experiments multiple times to ensure improved accuracy and repeatability. In Fig. 16 (c), the frequency shows very good agreement between the experimental and numerical results. The small discrepancy observed in the amplitude associated with damping may be attributed to dynamic resonance effects, experimental uncertainties, and factors not fully captured in the numerical model.

Transverse displacement contours of the solid shell configuration 2 under 200 Hz excitation at different time instants are shown in Figs. 17(a) through 17(d). These results depict the propagation of wave, showing the progression of the wavefront and its interaction with the free boundary.

Next, solid shell configuration 3 is considered with length of 84.52 mm, span of 50 mm, thickness of 4 mm, radius of curvature of 100 mm, and Rayleigh damping coefficient of $\alpha = 461.235\ \mathrm{s}^{-1}$. The shell is subjected to excitation frequencies of 200 Hz and 800 Hz, as depicted in Figs. 18(a) and 18(b), respectively. Transverse displacements are recorded at two locations: the input point (I), positioned at the center of the loading edge, and the output point (O_1), located 5 mm from the free edge along the y-axis, as shown in Fig. 18(c). Numerical analyses

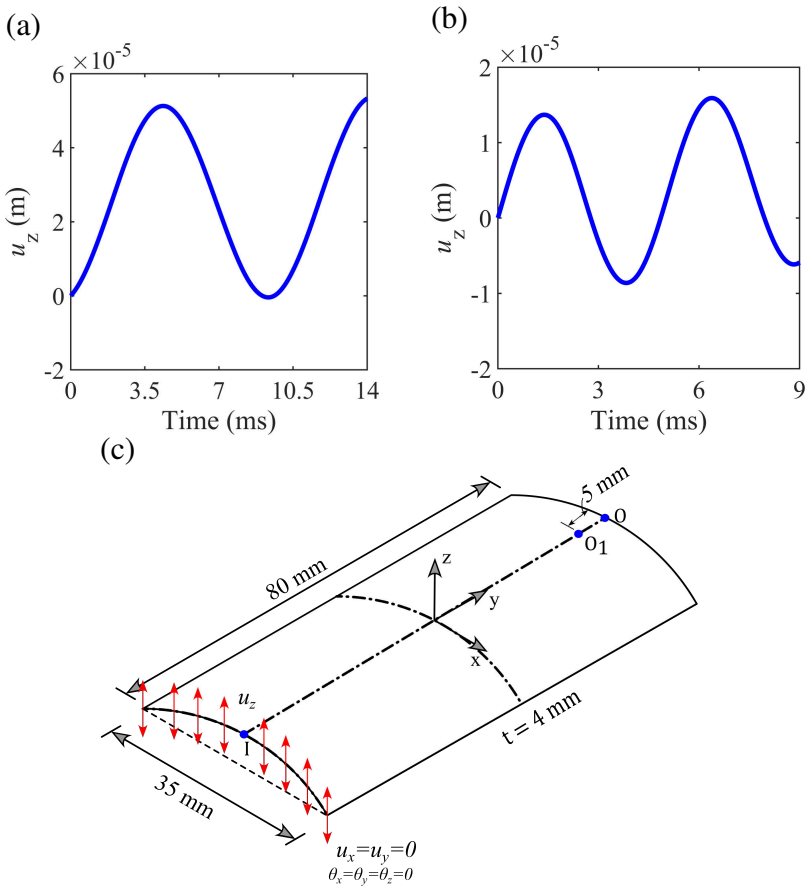


Fig. 15: Experimental transverse displacement inputs used in the solid shell configuration 2 simulations at different excitation frequencies: (a) 100 Hz, and (b) 200 Hz. (c) Reference configuration and boundary conditions of the solid shell configuration 2.

are carried out using PD model with 12665 material points and FEM model comprising 12556 elements, both subjected to displacement-controlled transverse loading.

The displacement time histories for both PD and FEM models, considering damping, are compared at location O_1[18], alongside experimental data for both 200 Hz and 800 Hz excitation frequencies, as shown in Figs. 19(a) through 19(d). Under 200 Hz loading, the peak displacement amplitudes from the experiment, FEM, and PD are as follows: 3.2799×10^{-5} m, 2.085×10^{-5} m, and 2.0133×10^{-5} m, respectively. Similarly, for 800 Hz excitation, the corresponding peak values are given by 2.2416×10^{-6} m, 1.5862×10^{-6} m, and 1.8132×10^{-6} m. Phase portraits comparing the experimental input, experimental output, PD output, and FEM output at excitation frequencies of 200 Hz and 800 Hz are presented in Figs. 19(e) and 19(f), respectively. For both the loading frequencies, the PD and FEM predictions closely match with the experimental result. The increasing trend of displacement with time in Fig. 19(c) may be attributed to small rigid-body motion of the plate during excitation with the electrodynamic shaker. To verify the consistency and reliability of the measurements, the experiments were repeated multiple times. The observed drift in the experimental response is mainly due to the presence of a very low-frequency rigid-body mode, which is difficult to completely eliminate in practical experimental setups and is not captured in the idealized numerical model.

Transverse displacement contours of the solid shell configuration 3 under 200 Hz excitation at different time instants are presented in Figs. 20(a) through 20(d). These results capture the wave propagation, show the advancement of wavefront, and its interaction with the free edge

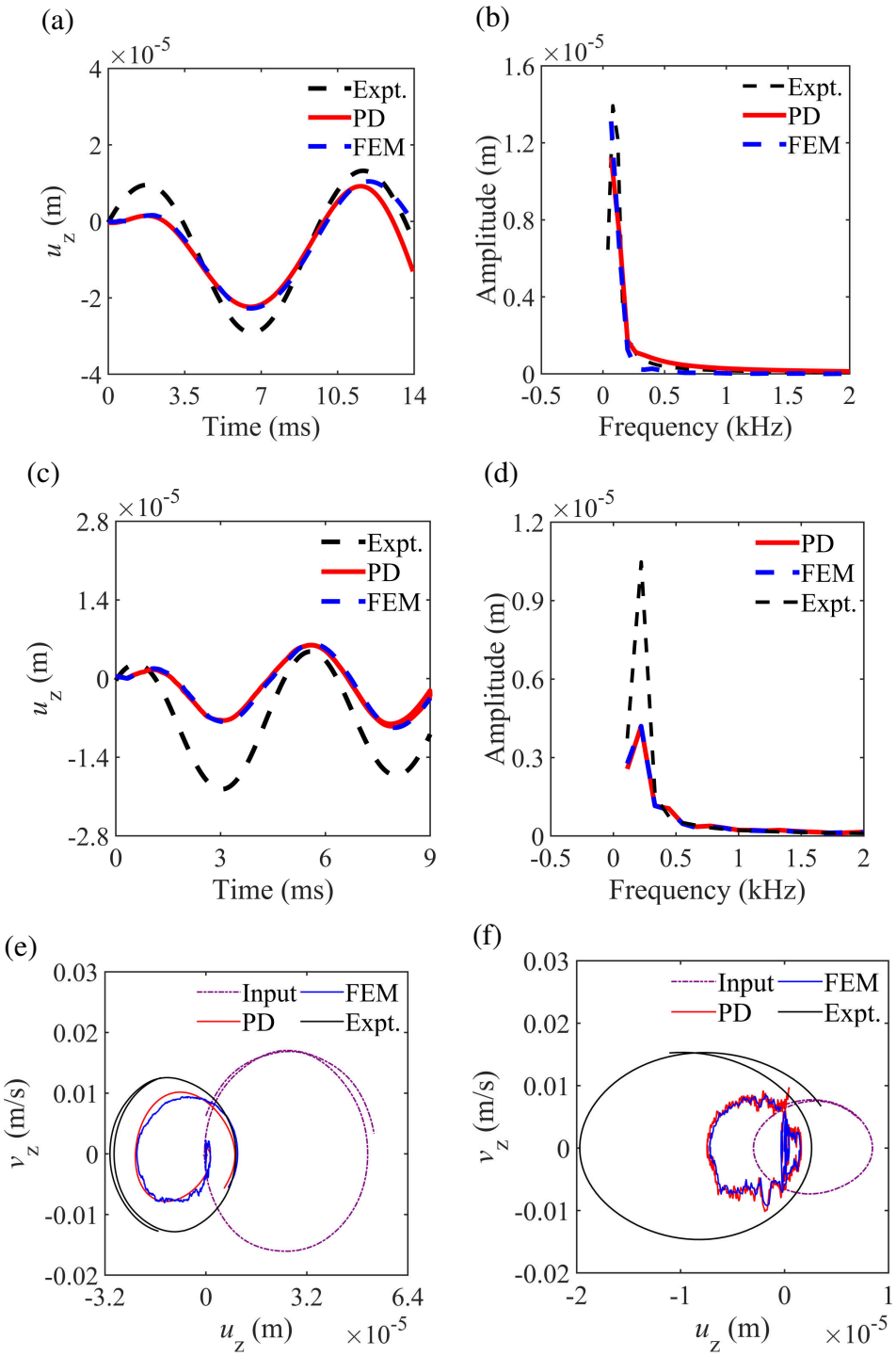


Fig. 16: Transverse displacement response (m) of TPU solid shell configuration 2 with damping. Comparison of experiment, PD, and FEM results: at 100 Hz excitation: (a) time history and (b) amplitude spectrum; at 200 Hz excitation: (c) time history and (d) amplitude spectrum. Phase portraits illustrating the response at excitation frequencies of (e) 100 Hz and (f) 200 Hz.

of the shell.

6.3. Mechanical metasurfaces

This section presents three mechanical metasurface configurations of plates and shells considering damping to investigate the effect of geometrical imperfections, viz., holes and cracks,

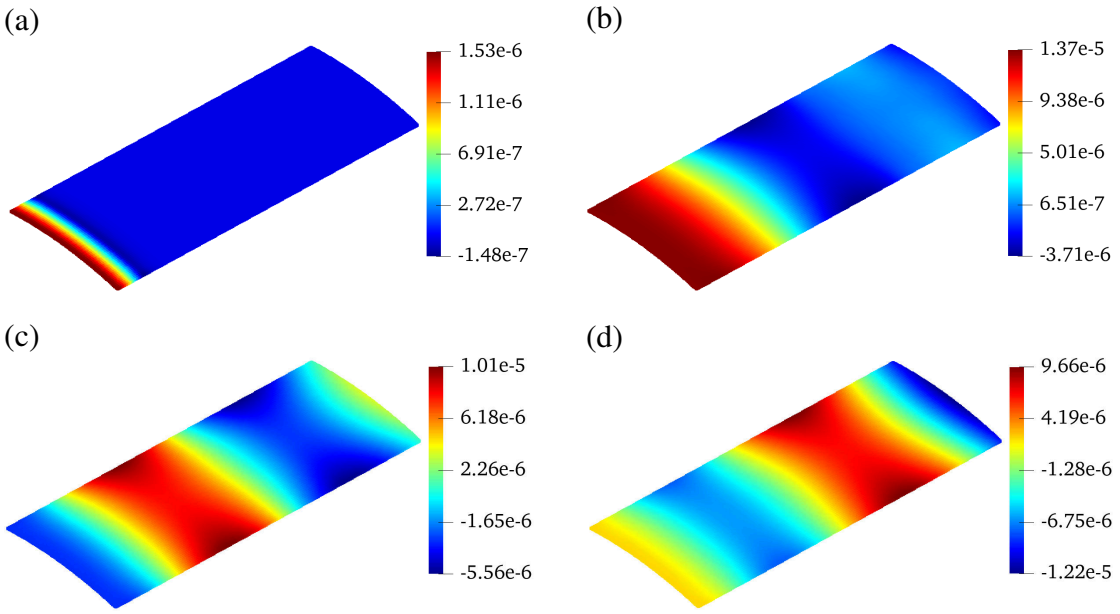


Fig. 17: Transverse displacement u_z contours (m) for TPU solid shell configuration 2, under 200 Hz excitation at times: (a) t = 0.1 ms, (b) t = 1.5 ms, (c) t = 3 ms, and (d) t = 5 ms.

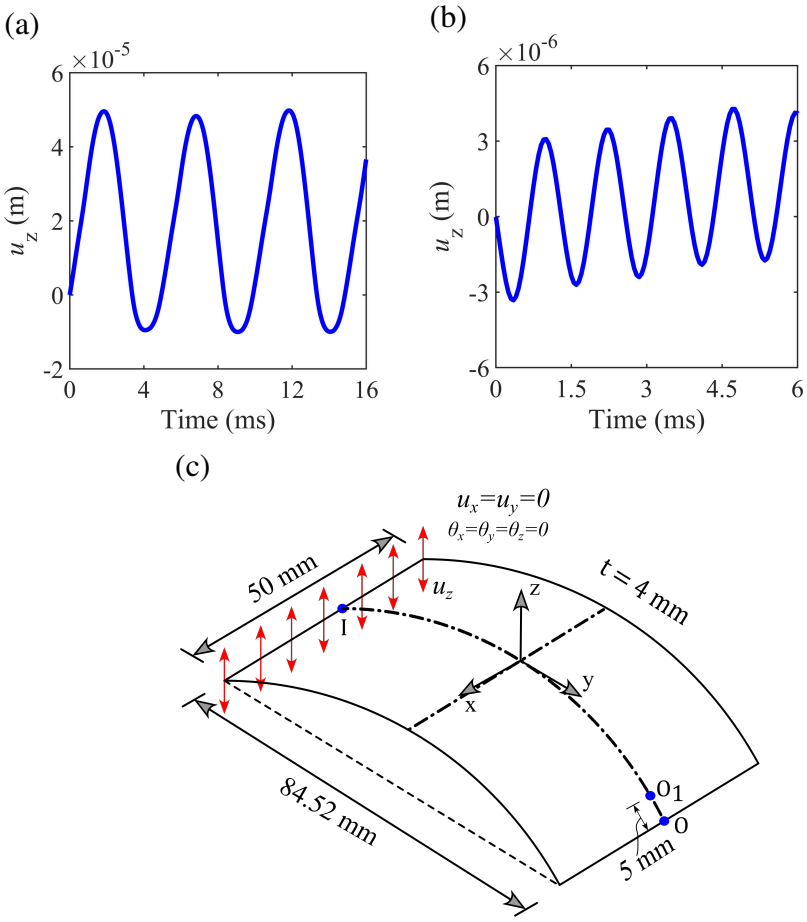


Fig. 18: Experimental transverse displacement inputs used in the simulation of the shell configuration 3 at different excitation frequencies: (a) 200 Hz, and (b) 800 Hz. (c) Reference configuration and boundary conditions of the solid shell configuration 3.

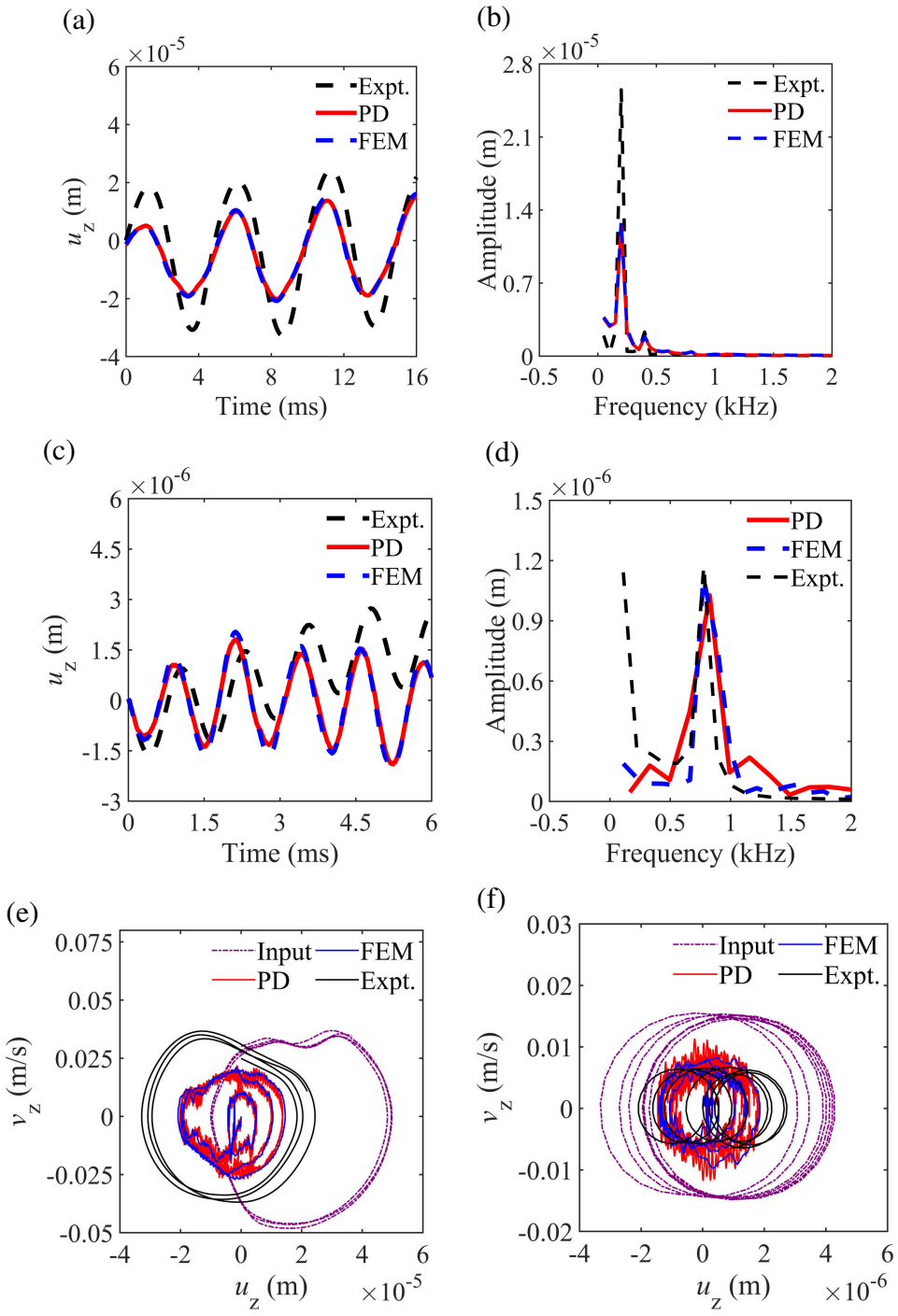


Fig. 19: Transverse displacement response (m) of TPU solid shell configuration 3 with damping. Comparison of experimental, PD, and FEM results: at 200 Hz excitation: (a) time history and (b) amplitude spectrum; at 800 Hz excitation: (c) time history and (d) amplitude spectrum. Phase portraits under transverse excitation at (e) 200 Hz and (f) 800 Hz.

on wave propagation. Mechanical metasurfaces are created by introducing periodically arranged holes and cracks in plates and shells to manipulate wave propagation. In shells with periodic circular holes, the cell walls among holes undergo large deformation and often buckling. For shells with periodic cracks, reflection of the elastic wave takes place at the crack surfaces, which changes the direction of the wave.

A TPU plate containing 24 circular holes, a TPU plate with 75 horizontal and vertical

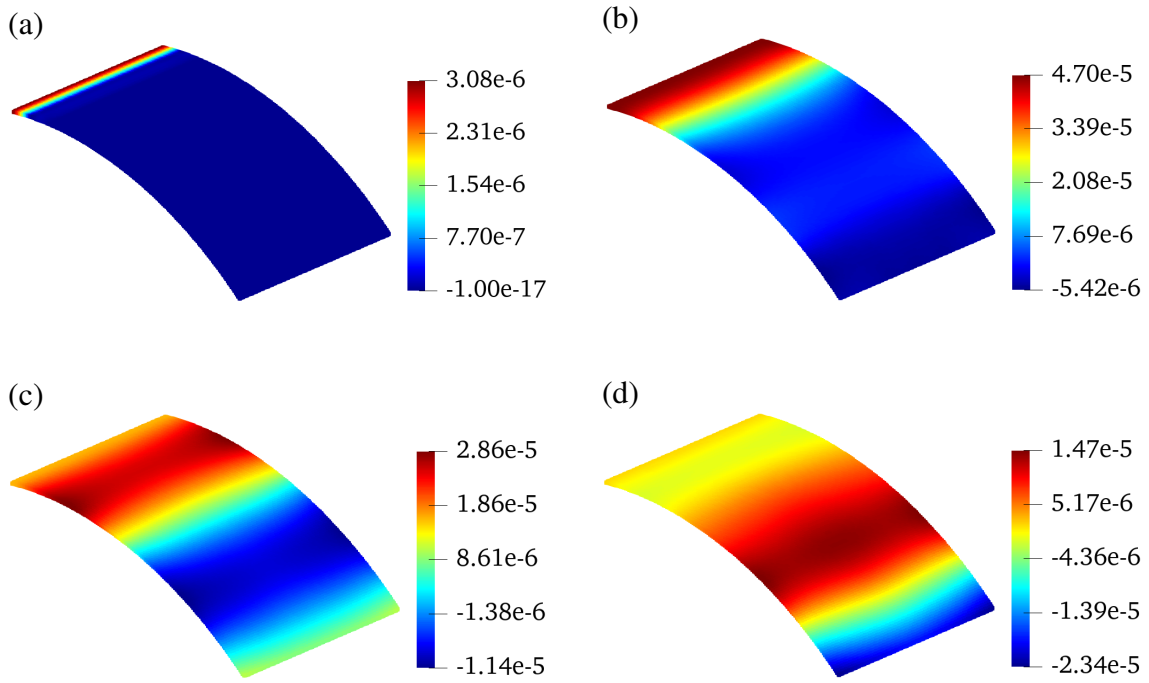


Fig. 20: Transverse displacement u_z contours (m) for TPU solid shell configuration 3, under 200 Hz excitation at times: (a) t = 0.1 ms, (b) t = 1.5 ms, (c) t = 3 ms, and (d) t = 5 ms.

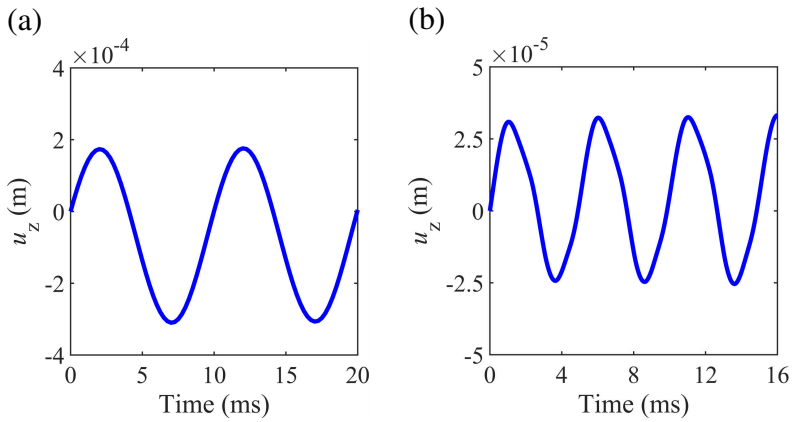


Fig. 21: Experimental input transverse displacement time histories for (a) a plate with multi-holes at 100 Hz and (b) a plate with multi-cracks at 200 Hz.

alternating cracks, and two SBR shells containing 25 periodic cracks and 25 periodic circular holes are considered. Displacements are measured at two locations, viz., the input point (I) located at the midpoint of the loading edge, and the output point (O_1 for plate and O for shell), positioned along the y-axis at a distance of 5 mm from the fixed edge for both the TPU plate and 7.6 mm for both the SBR shell, as indicated in Fig. 22. The effectiveness of each mechanical metasurface is evaluated by calculating transmittance in the frequency domain as

$$\text{Transmittance} = 20\log_{10}\left|\frac{\text{Amplitude of output displacement}}{\text{Amplitude of input displacement}}\right|. \tag{52}$$

The TPU plate has a length of 115 mm, width of 40 mm, thickness of 4 mm, and radius of curvature of 1000 mm. In the first configuration, the plate has circular holes with diameter of 5 mm, arranged at a center-to-center spacing of 13.33 mm, and the transverse edge-to-hole

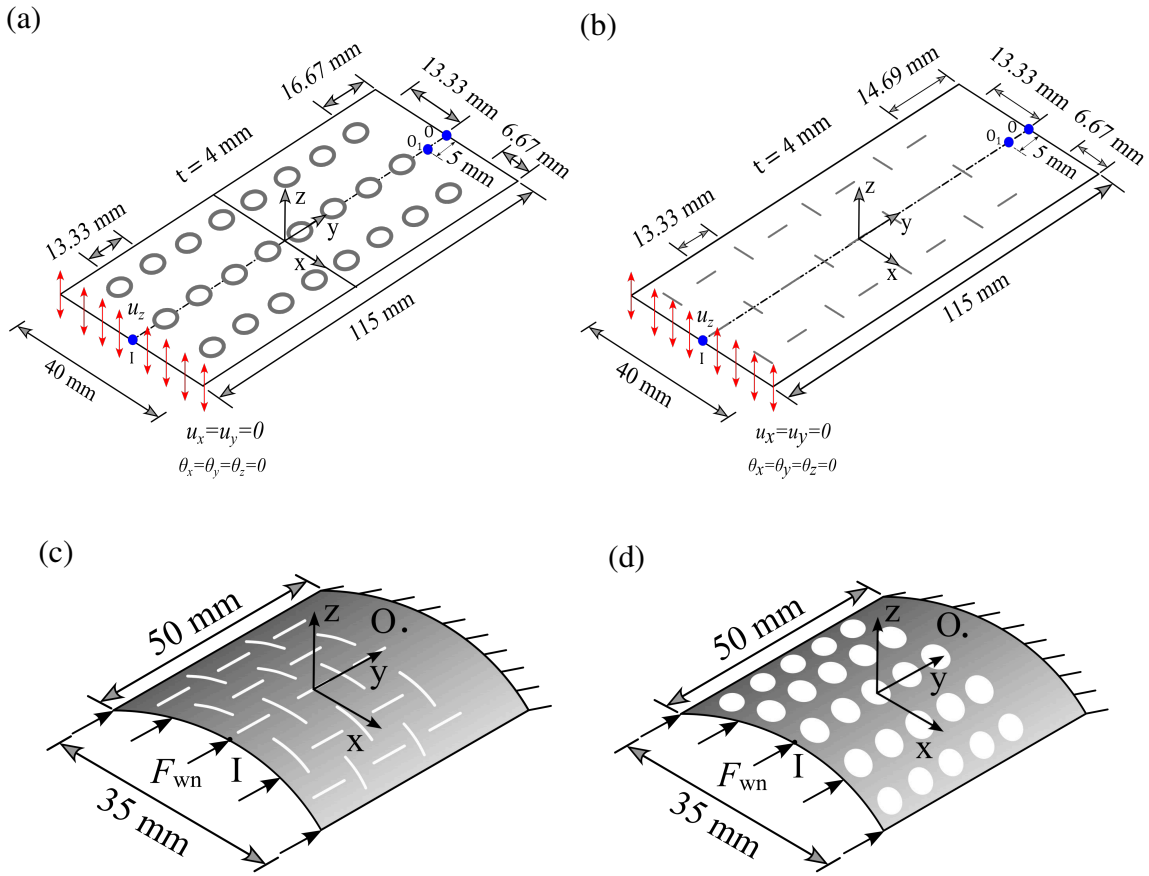


Fig. 22: Reference configurations of mechanical metasurfaces with boundary conditions: (a) TPU plate with periodically arranged circular holes, (b) TPU plate with periodic cracks, (c) cylindrical SBR shell with periodic cracks, and (d) cylindrical SBR shell with periodic circular holes.

center offset is 6.67 mm on each side as shown in Fig. 22(a). The PD model with 12815 material points and FEM model with 14647 elements are analyzed for displacement-controlled transverse loading at 100 Hz excitation [Fig. 21(a)]. In the second configuration, the TPU plate incorporates periodic cracks with a length of 5 mm and a width of 0.5 mm, arranged with a center-to-center spacing of 13.33 mm. The transverse edge-to-crack offset is 6.67 mm on each side, as illustrated in Fig. 22(b). For this case, the PD model consists of 7293 material points, while the FEM model includes 8148 elements, and both are subjected to displacement-controlled transverse loading at an excitation frequency of 200 Hz [Fig. 21(b)].

Transverse displacement response from PD and FEM in the presence of damping shows good agreement with experimental result in both time and frequency domains as shown in Figs. 23(a) through 23(d). The peak displacement amplitudes from the experiment, FEM, and PD analyses for plate with holes are given by 2.0680×10^{-4} m, 1.1464×10^{-4} m, and 1.3157×10^{-4} m, respectively. For the plate with periodic cracks, the corresponding peak displacement amplitudes obtained from the experimental, FEM, and PD analyses are 3.8243×10^{-5} m, 2.0271×10^{-5} m, and 2.0289×10^{-5} m, respectively. A comparison of the experimental input, experimental output, PD output, and FEM output for the TPU plate with holes at an excitation frequency of 100 Hz and for the TPU plate with cracks at 200 Hz are presented in Figs. 23(e) and 23(f).

Dynamic analyses of SBR mechanical metasurfaces are also performed. The shells have length of 50 mm, span of 35 mm, thickness of 1 mm, and radius of curvature of 100 mm.

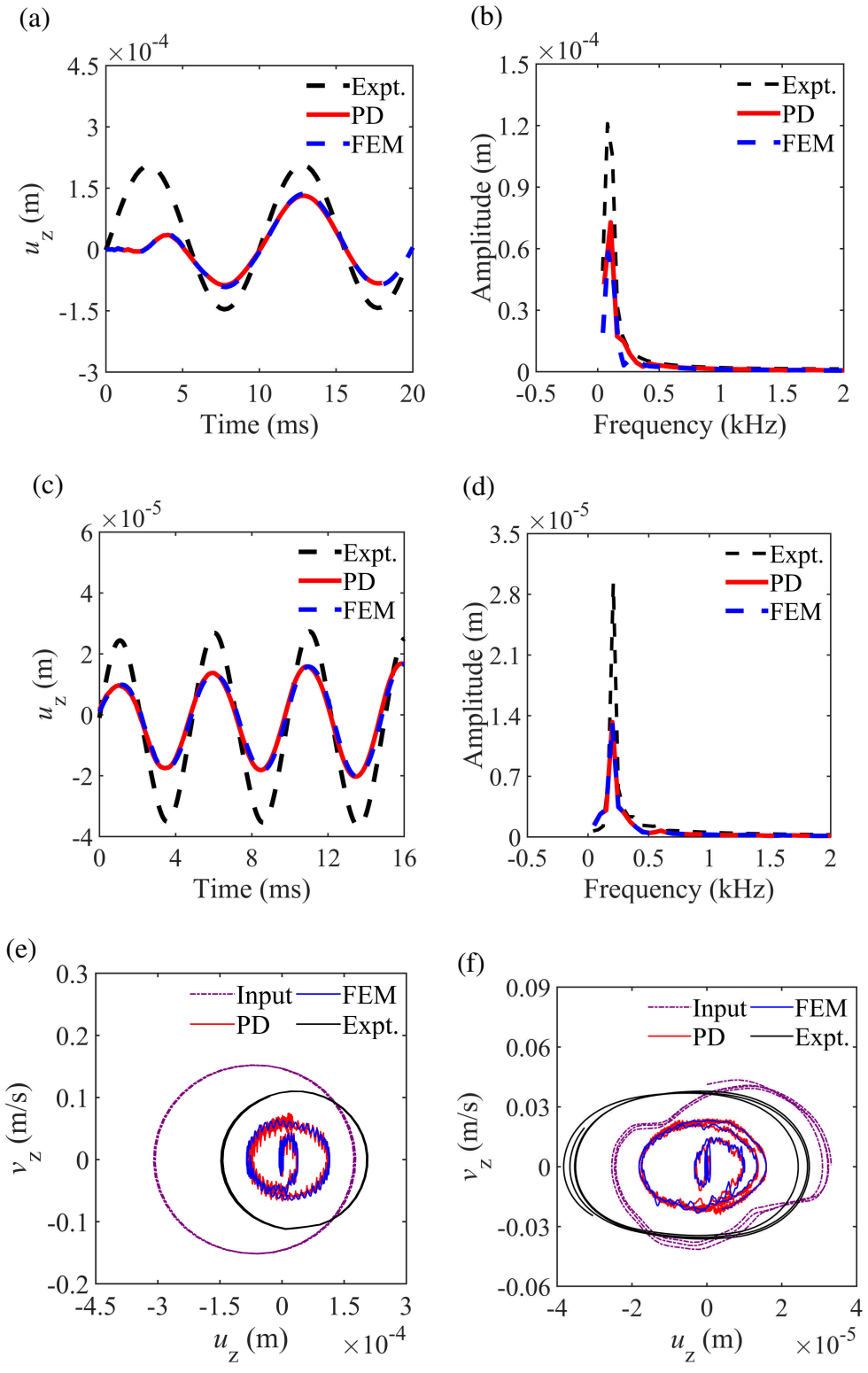


Fig. 23: Comparison of PD, FEM and experimental transverse displacement response for the TPU plate with periodically arranged circular holes under 100 Hz excitation with damping: (a) displacement time history and (b) amplitude spectrum. TPU plate with periodic cracks under 200 Hz excitation with damping: (c) displacement time history and (d) amplitude spectrum. Phase portraits of (e) plate with circular holes and (f) plate with cracks.

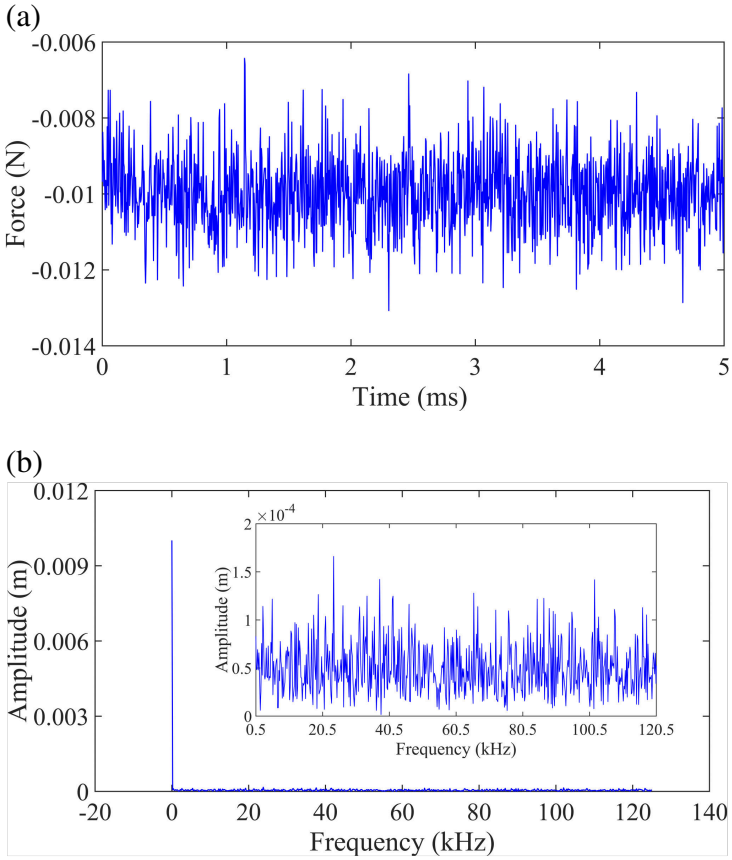


Fig. 24: (a) Time history of force generated as a realization of Gaussian white-noise with mean $F_{\text{wn}} = 0$ N and standard deviation $\sigma = 0.001$ N superimposed on a constant compressive force of magnitude 0.01 N and (b) force amplitude spectrum.

Material properties are considered as Young's modulus $E = 0.778$ MPa, Poisson's ratio $\nu = 0.45$, density $\rho = 980$ kg/m^3, and Rayleigh damping coefficient $\alpha = 409.5895$ s^{-1}. One curved edge of the shell is fixed. A force time history is considered where a realization of Gaussian white noise with mean $F_{\text{wn}} = 0$ N and standard deviation of 0.001 N is superimposed on a constant compressive force of magnitude 0.01 N. The force is applied on the opposite edge as shown in Figs. 22(c) and 22(d). The force time history considered in the simulation is furnished in Fig. 24(a) and force amplitude spectrum is shown in Fig. 24(b). Note that the constant force corresponds to zero-frequency in the force amplitude spectrum.

For the mechanical metasurface with holes, the Euclidean radius of each hole is 2.742 mm, with a center-to-center spacing of 6.784 mm with the nearest hole center 4.042 mm from both the loading and longitudinal edges. The configuration is discretized into 41162 particles. For the mechanical metasurface with cracks, each crack has a Euclidean length of 6 mm, and center-to-center spacing of 5.143 mm. Cracks are oriented orthogonally in alternating layers, with the nearest crack center positioned 7.324 mm from the loading and longitudinal edges. The shell geometry is discretized into 16129 particles.

The displacement at point O_1 for both plates [Fig. 22(a) and 22(b)] remains significantly lower than its value at loading point I. Figs. 25(a) through 25(d) show the displacement contours for plate with circular holes and Figs. 25(e) through 25(h) present the displacement contours of the plate with cracks, along the loading direction at various time points.

Similarly, in Figs. 22(c) and 22(d), the displacement at point O on the shell remains notably less than its value at the loading point I. Figs. 26(a) through 26(h) provide the evolution of axial

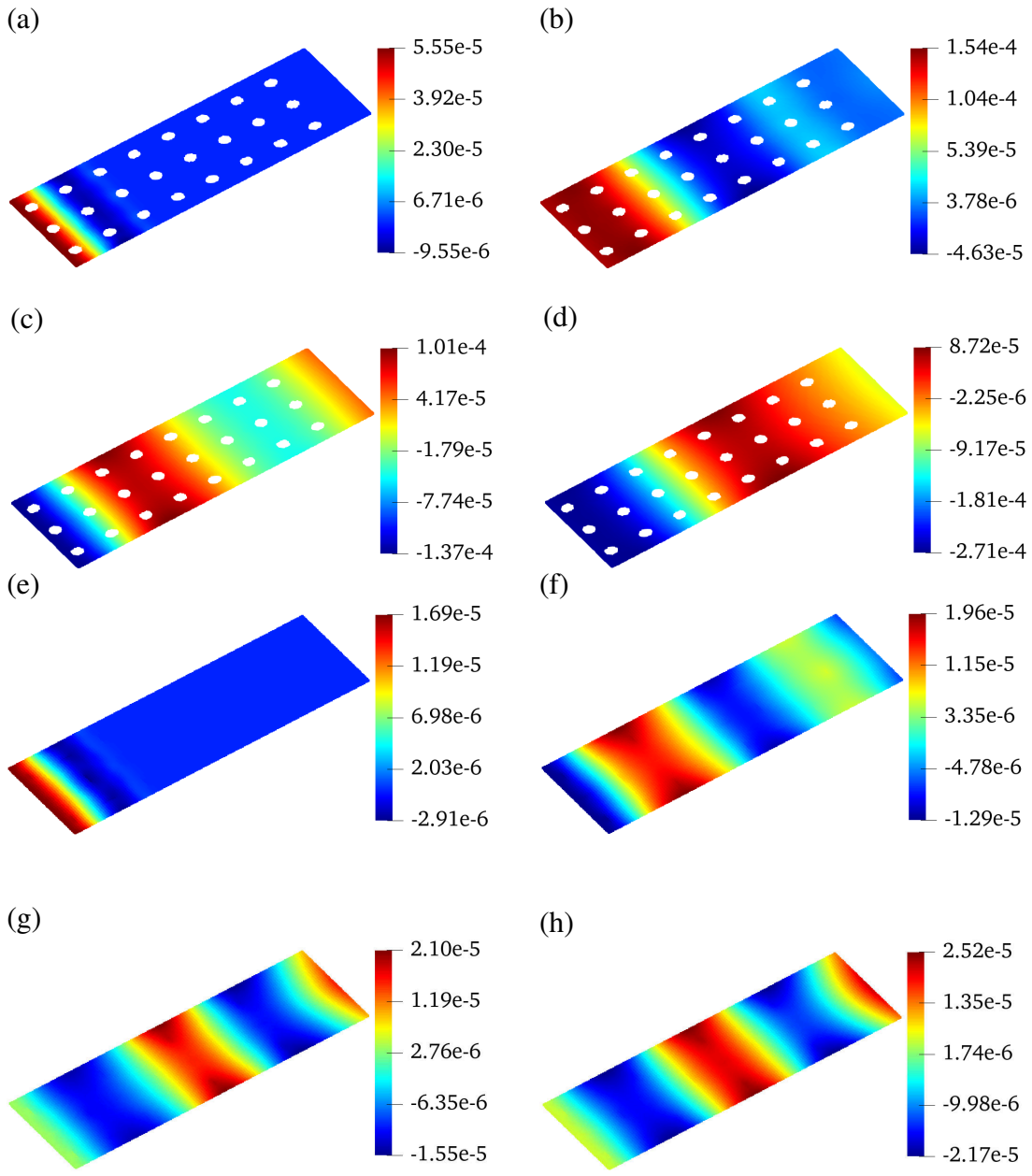


Fig. 25: Displacement contours (m) for mechanical metasurface configurations: contour of u_z (m) for TPU plate with periodic circular holes subjected to 100 Hz excitation at times: (a) t = 0.4 ms, (b) t = 2.8 ms, (c) t = 5 ms, and (d) t = 8 ms. For TPU plate with periodic cracks subjected to 200 Hz excitation at times: (a) t = 0.4 ms, (b) t = 3 ms, (c) t = 5 ms, and (d) t = 10 ms.

displacement (u_y) over time.

The bandgap is the range of frequencies for which the output displacement is smaller than the input displacement. Fourier amplitude spectra at input (I) and output (O) points show reduced displacement amplitudes at the output. The comparison of the input and FEM output transverse displacements for the TPU solid plate [Fig. 7(c)] and the TPU plates with periodic holes or periodic cracks is shown in Figs. 27(a) and 27(b), respectively. For the plate with holes, a 100 Hz [Fig. 21(a)] input is applied to both the metamaterial plate and a solid plate of the same dimensions, while for the plate with cracks, a 200 Hz [Fig. 21(b)] input is used for both the metamaterial and corresponding solid plate. In both cases, the metamaterial plates exhibit a reduction in amplitude compared to the corresponding solid plates, demonstrating

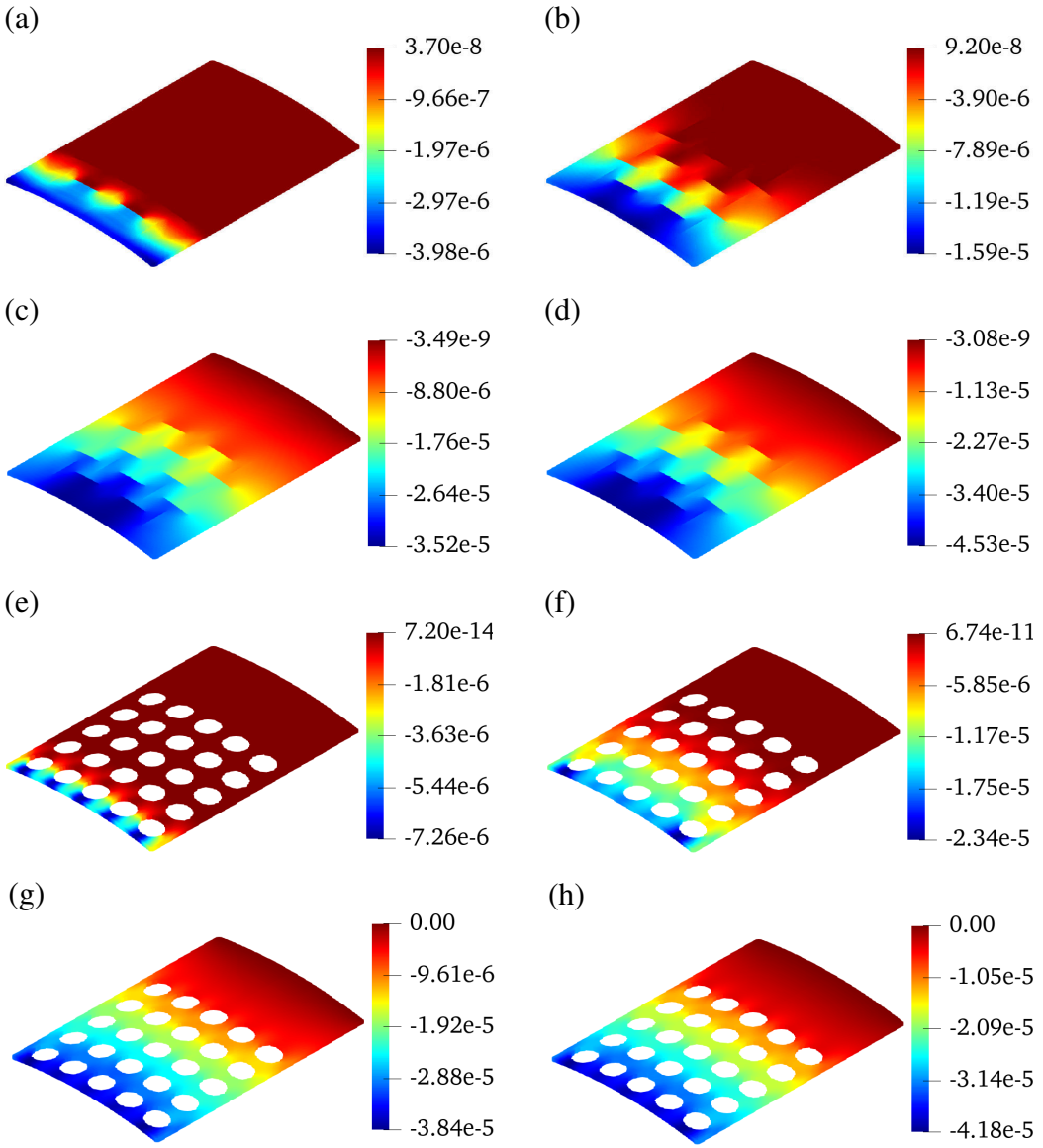


Fig. 26: Displacement contours (m) for mechanical metasurface configurations: contour of u_y (m) for SBR shell with periodic cracks at times: (a) t = 0.4 ms, (b) t = 1.2 ms, (c) t = 2.8 ms, and (d) t = 4.8 ms; and contour of u_y (m) for SBR shell with periodic circular holes: (e) t = 0.4 ms, (f) t = 1.2 ms, (g) t = 2.8 ms, and (h) t = 4.8 ms.

wave attenuation and highlighting their effectiveness. Transmittance plots shown in Figs. 27(c) and 27(d) confirm significant wave amplitude reduction and bandgap formation for the proposed mechanical metasurface designs, where crack-based mechanical metasurface exhibits best wave isolation behavior. Phase portraits for both SBR shell configurations are shown in Figs. 27(e) and 27(f). Geometric nonlinearity is considered in both PD and FEM simulations for all the problems. Note that although displacement values are small in our simulations, the displacement gradients and strains are not small, as during wave propagation, displacement values change spatially. Therefore, in this case, we cannot neglect the nonlinear term in the Green-Lagrange strain tensor as in the small deformation approximation.

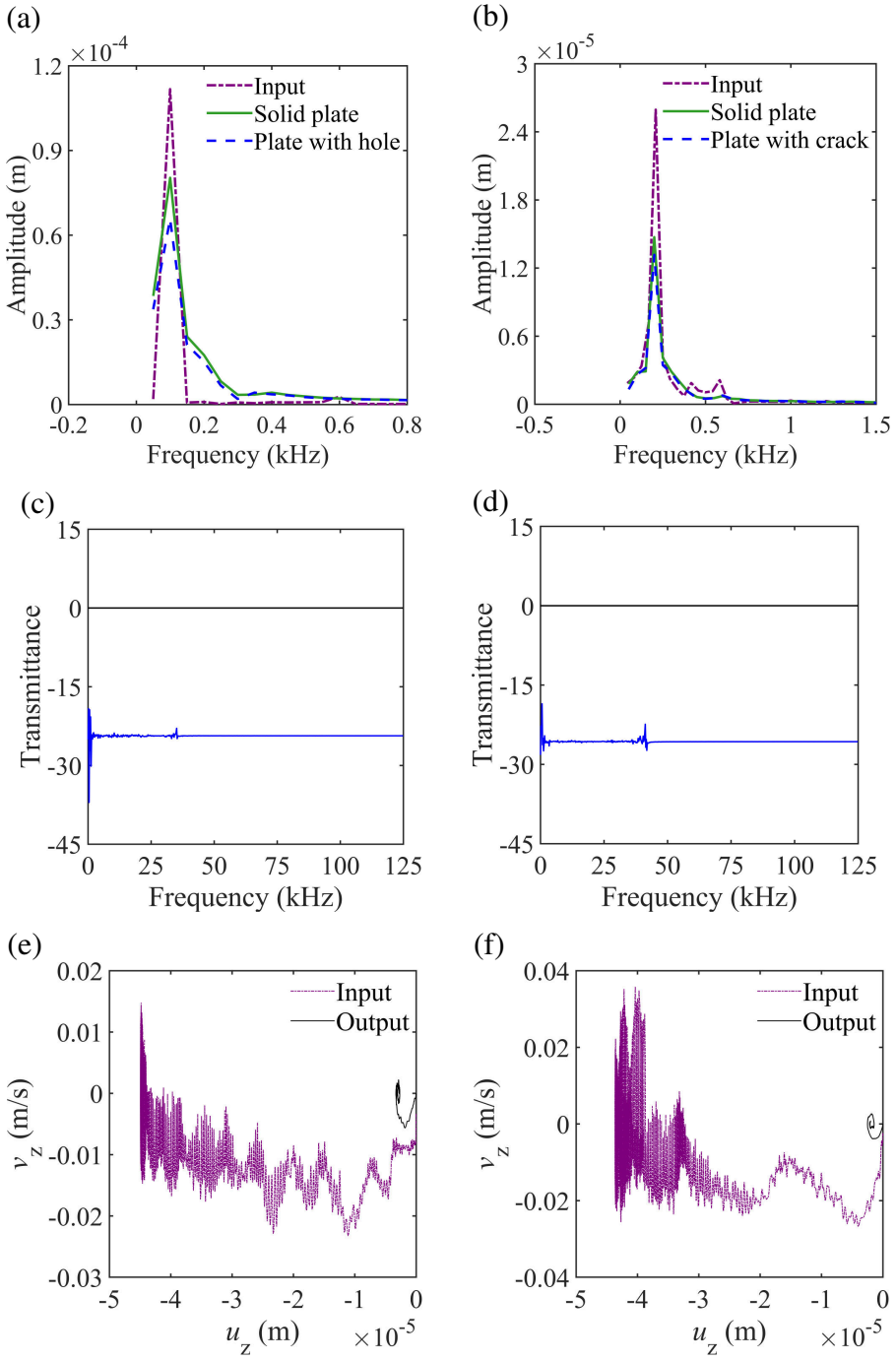


Fig. 27: Comparison of transverse displacement response at input and output points of TPU solid plate and TPU plate with holes or cracks under identical loading conditions: (a) a solid plate and a plate with periodic circular holes under 100 Hz excitation; (b) a solid plate and a plate with periodic cracks under 200 Hz excitation.; transmittance spectra for mechanical metasurface configurations of (c) cylindrical SBR shell with periodic cracks and (d) cylindrical SBR shell with periodic circular holes. Phase portraits for (e) the shell with cracks and (f) the shell with circular holes.

7. Conclusions

This work presents a finite deformation Simo-Reissner NOSB-PD shell theory extended to include damping for analyzing wave propagation and wave isolation in shells and mechanical metasurfaces. By introducing damping into the PD shell governing equations, the proposed

framework successfully captures energy dissipation, which is typical in materials like polymer. The governing equations are solved using the Newmark-beta method in conjunction with the Newton-Raphson method, ensuring numerical stability and convergence for dynamic problems. Validation exercises on TPU plates and shells under harmonic excitation demonstrate agreement between PD predictions, FEM solutions, and experimental results in both time and frequency domains.

Further investigations on mechanical metasurface configurations including TPU plates with periodic holes, periodic cracks, and SBR cylindrical shells with periodic cracks and holes highlight the effectiveness of these geometric imperfections in reducing wave propagation through mechanisms such as large deformation and wave reflection. These findings demonstrate the capability of the proposed PD shell theory with damping to model wave isolation response of engineered mechanical metasurfaces. These metasurfaces can be used to protect certain important parts of the structure from the incident wave-induced damage. This work may be extended to analyze wave propagation through mechanical metasurface with inclusions and local curvature.

CRediT authorship contribution statement

Nilesh Choudhary: Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing - original draft, Writing - review & editing. **Tanuj Gupta:** Data curation, Investigation, Methodology, Software, Validation, Visualization, Writing - original draft, Writing - review & editing. **Ankur Dwivedi:** Investigation, Methodology, Software, Validation, Visualization, Writing - original draft, Writing - review & editing. **Bijaya Samal:** Investigation, Methodology, Software, Validation, Visualization, Writing - original draft. **Pranesh Roy:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Writing - original draft, Writing - review & editing. **Rahul Bhartiya:** Funding acquisition, Project administration, Resources, Supervision, Writing - review & editing. **Bishakh Bhattacharya:** Conceptualization, Project administration, Resources, Supervision, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Dynamic crack propagation in cylindrical shell

The advantage of PD modeling over classical continuum mechanics lies in its ability to model discontinuities such as cracks. This makes PD particularly suitable for problems where crack initiation and propagation are the primary focus. In the case of stationary crack, linear elastic fracture mechanics predicts stress singularity at the crack tip. In FEM, this stress

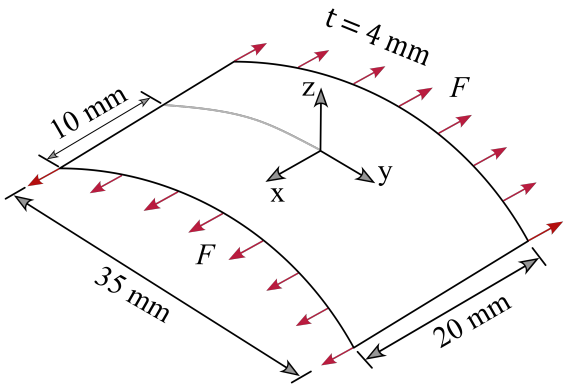


Fig. A.1: Reference configuration and boundary conditions of the cylindrical shell with edge notch.

singularity results in divergence of solution with mesh refinement. Also, PD is free of mesh distortion unlike FEM due to the meshfree nature of its discretized form. In order to demonstrate crack propagation, we added a new simulation of dynamic crack propagation in this section.

The analysis is performed on a cylindrical shell having a span of 35 mm, a length of 20 mm, a thickness of 4 mm, and a radius of curvature of 100 mm, as illustrated in Fig. A.1. The edge notch extends up to 17.5 mm along the y-direction, corresponding to half of the shell span. In this study, SBR material is used in the analysis with a critical stretch of 0.3432 and a critical relative rotation of 3.5×10^{-2}.

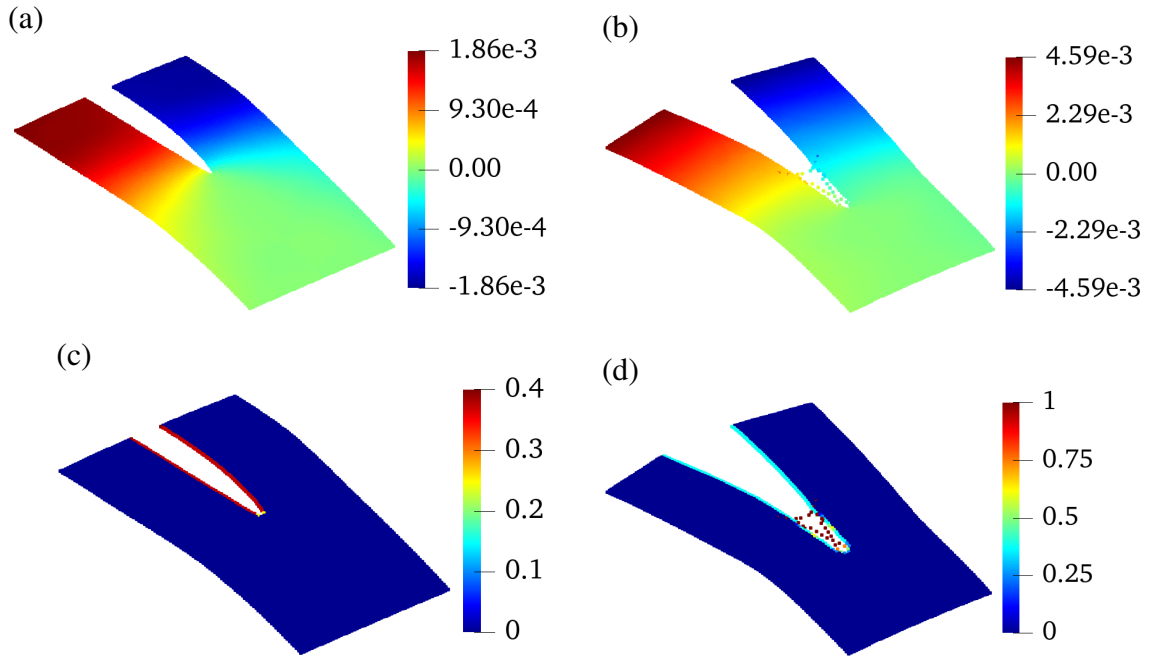


Fig. A.2: Displacement contour plots (m) at times (a) t = 1.25 ms and (b) t = 1.85 ms, and damage contour plots at times (c) t = 1.25 ms and (d) t = 1.85 ms for the cylindrical shell with an edge notch.

The cylindrical shell is discretized into 5600 particles for PD simulation. A predictor–corrector time integration scheme is employed to determine the transient response, with a time step of 5×10^{-9} s to ensure numerical stability and accuracy. A uniformly distributed tensile force of total magnitude F = 3 N is applied symmetrically along both curved edges of the cylindrical shell, as illustrated in Fig. A.1.

The displacement and damage contours with the advancement of time are presented in the

Fig. A.2. The contours show that crack initiates and propagates due to the interaction of wave with the crack. Also, the shell undergoes large deformation during this fracture process. This highlights the necessity of considering finite deformation shell model in PD framework.

Data and code availability

Data will be made available on request.

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